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# Sticky Spending, Sequestration, and Government Debt†

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*Once established, government spending programs tend to continue. A commonly held view is that spending inertia leads to unsustainable debt, ultimately requiring fiscal adjustments such as "sequestration." We show that by insuring against political turnover, inertia may reduce politicians' incentives to accumulate debt. However, large preexisting commitments and the prospect of future stabilization can lead to overspending to dilute past administrations' commitments. Finally, we show that political polarization amplifies incentives to prioritize inertial programs, potentially explaining the increased share of mandatory spending in the US budget.* (*JEL* D72, E62, H61, H63)

Public spending is sticky: once new programs are established, they tend to persist over time. For instance, entitlement spending programs such as Social Security and Medicare are mandated by existing laws and continue on autopilot unless a new law is passed that alters them. Moreover, due to political and bureaucratic pressures, discretionary programs such as education and defense are often maintained despite automatically expiring at the end of the budget year, without any legal obligation to be renewed.[1]

A commonly held view is that inertia makes spending uncontrollable and may send the economy on an unsustainable fiscal trajectory, which ultimately requires fiscal adjustments to be made. The unsustainability of current policies appears to be the "fiscal problem of the 21st century" (Jones 2003).[2] According to many commentators, solving this problem would require implementing less inertial budgeting

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[1] Spending may become entrenched by changing voters' reference point (Alesina and Passarelli 2019 and Charité, Fisman, and Kuziemko 2015). Inertia may also result from bargaining frictions (e.g., Krehbiel 1998 and Osborne and Rubinstein 1990) or may be due to labor laws that make it difficult to transfer public employees across spending agencies.

[2] See CBO (2020). Auerbach, Gokhale, and Kotlikoff (1994); Bohn (2005); and Hamilton (2013) evaluate the present value of implicit commitments (e.g., Social Security and Medicare) to assess the sustainability of US fiscal policy.

practices.[3] Along these lines, OECD (2012) has recommended sunset clauses specifying a date beyond which programs automatically expire. Proposals to reduce budgetary inertia are currently being debated in US politics.[4]

This paper develops a political economy model to study the impact of budgetary inertia on spending and debt. Besides the mechanical effect of inertia, which results in increased spending due to the continuation of past programs, we show that persistence and the prospect of future stabilization change the *strategic* interactions across governments, significantly impacting spending incentives. Contrary to the common view, we show that with functioning checks and balances, such as a supermajority requirement, persistence can reduce the overaccumulation of debt and even restore efficiency. However, with weaker checks and balances, high inertia could exacerbate politicians' incentives to spend, speeding up debt accumulation and making future cuts even more painful. Moreover, in addition to higher debt, we show that political polarization leads to an overprovision of inertial programs compared to less inertial ones, helping to explain the increasing US budget share of mandatory spending.

We derive these findings by introducing spending inertia into a standard model of strategic debt. The current political economy paradigm, such as Tabellini and Alesina (1990); Battaglini and Coate (2008); and many others, has shown that politicians tend to overaccumulate debt. Typically, however, debt sustainability is not a concern. This is because the standard assumption is that budgets are not inertial; when each program must be justified afresh each year, there are, strictly speaking, no "fiscal adjustments" to be made to ensure sustainability. Introducing budgetary stickiness enables us to study not only whether the debt is inefficiently high but also whether the debt is on an unsustainable path.

We consider a simple setting to make our point clearly. Two goods are supplied by the government and financed by nondistortionary taxes and debt. Two parties stochastically alternate in power and disagree as to how to allocate the budget.[5] As is well-known, this setting predicts a "deficit bias," wherein the incumbent overspends on her preferred good and issues debt to "force" the subsequent government to spend less on the good she does not value. The standard approach is to assume that the incumbent can cut spending programs put in place by previous administrations. In the data, however, spending programs are highly persistent, and we observe limited shifts in budget allocations following government changes (see Section III). We thus assume that due to legal and political costs, the incumbent must maintain an *exogenous* fraction $\alpha \geq 0$ of past spending. The higher $\alpha$, the stickier the budget. We start by assuming that the same $\alpha$ applies to all programs, and then

[3] For example, Schick (2003, p. 21) writes, "Arguably, if budgeting were less incremental and more open to a review of base expenditures, governments and deficits would be smaller."

[4] A group of US senators has recently sponsored the Zero Based Budgeting Act (S.2463–117th Congress 2022), which requires the periodic review of all discretionary programs. Moreover, some US politicians have suggested that entitlements should be subject to annual appropriation bills (Tankersley, Jim. 2022. "Republicans, Eyeing Majority, Float Changes to Social Security and Medicare." *New York Times*, November 2.).

[5] In the United States, for instance, Democratic voters are more likely than Republicans to favor increasing funding for health care, education, environmental protection, and Social Security. Conversely, Republicans are more likely than Democrats to support increasing military defense, border control, and anti-terrorism spending (Pew Research Center 2019).

in Section VB we relax this assumption, introducing mandatory programs that are stickier than discretionary ones.

Budgetary inertia brings out the sustainability issue and the implied timing of adjustment. Throughout the main text of the paper, we assume that the budget must always be sustainable; as soon as the present value of spending commitments is larger than the present value of government revenues, a budget "sequestration" occurs, meaning that the incumbent cuts all programs proportionally and has no discretion to choose which entitlements to curtail.[6] In US law, sequestration refers to the implementation of automatic, across-the-board spending cuts. In this paper, we will use the term *sequestration* to indicate, more broadly, a fiscal consolidation. As shown in Section IIB, fiscal consolidations by a deliberate government decision typically involve spending cuts that follow a proportional pattern, similar to the US sequestration, so that fiscal adjustments generally maintain the preexisting ranking among spending programs in terms of size.

The current government can affect future policymakers in two ways. The first is through increased debt, which reduces future governments' fiscal space; this is the standard channel emphasized by the strategic debt literature. The second is through spending. When policies are sticky, spending becomes an additional state variable. Indeed, politicians do internalize that spending is persistent. For instance, in 2009 Republicans expressed concern that President Obama's stimulus package would become entrenched.[7] More recently, Senator John Thune opposed President Biden's recovery plan, stating that "[Democrats] believe that once they put these programs in place, no one from either party will be able to get rid of them."[8]

Sticky spending has two main strategic effects. First, by allocating funding to her preferred programs, the incumbent locks in future spending in case she is voted out of office, attenuating the spending fluctuations due to political turnover. This *insurance* motive has an ambiguous effect on spending and debt accumulation. On one hand, for small and strictly positive $\alpha$, as $\alpha$ increases, the present government can more effectively bind the choices of future governments. This strengthens the incentive to spend, validating the common view that budgetary inertia leads to large debt buildup. On the other hand, when $\alpha$ is sufficiently large, the incumbent attains full insurance. As the stickiness increases, the spending needed to lock in subsequent governments decreases, reducing the incentives to overspend and accumulate debt. As a result, as long as previous entitlements are not large, there is a hump-shaped relationship between spending persistence and debt accumulation.

When preexisting entitlements are sufficiently large, another mechanism comes into play: faced with the prospect of future stabilization, there is a *dilution* motive to spend. One might expect that the "threat" of fiscal cuts would discipline the current incumbent. We find instead that it could aggravate the incentives to overspend. Because each program is reduced *proportionally* upon sequestration, larger

[6] In online Appendix D, we will relax the assumption that the sequestration is triggered automatically; the incumbent may choose to delay it by running additional debt.

[7] According to Senator Lisa Murkowski, "The question then needs to be asked: Will this level of funding become the new baseline for the Department?" 155 *Congressional Records* S2278 (2009).

[8] 167 *Congressional Records* S7352 (2021). More colorfully, Senator Mitch McConnell stated that Biden's spending increase would shift the United States "from a temporary emergency to permanent socialism" (167 *Congressional Records* S6465, 2021).

programs will claim a bigger share of available resources. In anticipation of consolidation, the incumbent may then increase the size of their preferred programs to dilute the entitlements of the opposition in a future sequestration.[9] The dilution effect is present when the incumbent inherits large commitments on spending programs that she does not value. Moreover, this motive arises only when budgets are sticky; when $\alpha$ is zero, there are no government commitments to begin with, and politicians have no incentive to dilute past commitments.

Taking all these effects together, we show that inertia has a double-edged effect on spending incentives. In some cases, higher budgetary stickiness reduces the need to strategically use debt. However, if preexisting entitlements are large, as in the current US context, the dilution motive operates in full, and budgetary stickiness worsens the deficit bias, even beyond the standard "strategic-manipulation" effect.

An implication of this theory is that budgetary stickiness and the expectations of future cuts may fuel a spending spiral, in which loose fiscal policy by past governments triggers even looser policies by subsequent ones. This may be a good description of how the two major American parties have been flirting with fiscal unsustainability. During the Obama administration, Democrats approved a steep rise in spending. Facing the sharp rise in entitlements and frustrated by their failure to repeal Obamacare, Republicans pushed for a loose fiscal stance: funding the border wall, spending more on defense, and cutting taxes. When the opposition's entitlements cannot be cut directly, the dilution channel predicts that the incumbent will do it indirectly through sequestration. In Section VIC, we provide anecdotal evidence that strategic considerations in line with our model may have played a role in US politics. By examining several episodes of fiscal consolidations across OECD countries, we also show that consolidations are indeed preceded by a spending acceleration.

The previous results hinge on the assumption that spending programs are downwardly rigid, but the incumbent government has full discretion to increase spending. Can further checks and balances prevent inefficient outcomes? To answer this question, we assume that any spending increase must be negotiated with the opposition. This captures supermajority requirements needed to pass new legislation, such as in the US Senate. In contrast to the environment without checks and balances, we show that stickiness can eliminate debt accumulation, even when the opposition's entitlements are large. In addition, higher opposition entitlements raise the opposition's bargaining power and implement a more equitable spending mix. In recent years, many important US reforms (e.g., the Bush and Trump tax cuts and parts of Obamacare) were approved by bypassing the Senate filibuster through so-called budget reconciliation. Our results show that debt would have been lower without these simple-majority expedited measures.

Finally, we extend the model by assuming that some programs are not sticky; this captures discretionary spending, which requires annual appropriation bills and, thus, can be thought of as less sticky than mandatory spending. We show that the presence of discretionary spending does not qualitatively change our results but does expose

[9] This echoes the debt-dilution channel in the sovereign default literature. When a country approaches a financial crisis, its government may have an incentive to issue new bonds, which dilutes existing creditors (Bolton and Jeanne 2009; Chatterjee and Eyigungor 2012a; and Hatchondo et al. 2016).

new dimensions of inefficiency. Since mandatory spending can be used to manipulate future governments, it is overprovided compared to the first best. In Section VC, we also show that as the preference polarization between the two parties intensifies, the incentive to overprovide mandatory spending becomes stronger. Our model thus implies that increasing political polarization can account for two potentially distinct phenomena: higher debt as well as a shift in the composition of government spending away from nonsticky (discretionary) programs. Along these lines, we present evidence in Section VIC showing a co-movement between increasing polarization, rising debt, and a growing share of mandatory spending in the budget. In Section VIC, we also demonstrate that increasing polarization is associated with a decrease in the budget share of "neutral" spending, i.e., spending not "owned" by a party. Applying similar reasoning as before, one possible explanation is that as polarization increases, bipartisan spending is less valuable in terms of "strategic manipulation" since both parties equally value it.

The rest of the paper is as follows. In Section I, we review the literature. In Sections II and III, we provide evidence to justify our modeling assumptions. In Section IV, we present the benchmark model. In Section VA, we consider a bargaining model in which the incumbent needs the opposition's approval to increase spending. In Section VB, we allow for two types of spending with different degrees of stickiness. In Section VC, we discuss a model with less than full polarization. Section VI examines some empirical implications of our model, and Section VII concludes. All proofs are in the Appendix.

## I. Literature Review

Our paper builds on the political economy literature on strategic debt.[10] A key difference between that literature and our paper is that we introduce budgetary inertia, which allows us to investigate debt sustainability and delayed stabilization.

Bouton, Lizzeri, and Persico (2020) were the first to study how entitlements affect debt accumulation. Their paper is related to ours, but our approaches differ. They model entitlements as transfers that are decided one period in advance; the incumbent chooses the current public good and the entitlements that her constituency will receive in the next period. We assume instead that current spending and entitlements are not two separate decisions; future entitlements can be created only by spending in the current period, thus capturing the idea that budgets are inertial. An important consequence of this assumption is that entitlements are potentially unsustainable in our model. In their model, instead, the incumbent must satisfy an intertemporal budget constraint when choosing future entitlements. Moreover, in their paper, entitlements and debt are substitutes; entitlements allow the incumbent to constrain future governments, which weakens the incentive to use debt. In our model, entitlements can be created only by spending and, potentially, running debt. Because of this, budgetary stickiness may contribute to high debt buildups.

[10] See Persson and Svensson (1989); Tabellini and Alesina (1990); Alesina and Tabellini (1990); Lockwood, Philippopouslos, and Snell (1996); Lizzeri (1999); Battaglini and Coate (2008); Yared (2010); and Debortoli and Nunes (2013). For comprehensive surveys of the literature, see Alesina and Passalacqua (2015) and Yared (2019).

Our work is also related to a growing political economy literature that studies policy inertia. In the "endogenous status quo" literature, policy inertia arises endogenously in a dynamic legislative-bargaining model (see the overview by Eraslan, Evdokimov, and Zapal 2020). The key assumption is that the default option in case of disagreement coincides with the previous period's policy (i.e., the status quo). A result of this literature (e.g., Baron 1996; Bowen, Chen, and Eraslan 2014; and Azzimonti, Karpuska, and Mihalache 2023) is that the endogenous status quo exerts an important insurance effect by raising the bargaining power of the non-proposing players.[11] Similarly to these papers, we find that sticky spending provides insurance. The key difference between the prior literature and our paper is that the former does not study debt, which is our focus. Moreover, by introducing debt-sustainability issues, we highlight the dilution channel, which has not yet been studied by this literature. Another strand of the political economy literature studies policy inertia in models without bargaining but with a cost of changing policy (e.g., Gersbach, Muller, and Tejada 2019; Gersbach et al. 2023; Eraslan and Piazza 2020; and Chatterjee and Eyigungor 2020). These papers, which also abstain from studying debt, find conditions under which costly policy changes lead to lower policy fluctuations. Our approach differs, in that we assume an asymmetric cost; in our baseline model, spending is downwardly rigid but can be increased at no cost.

Our paper is also related to the influential work by Alesina and Drazen (1991) on delayed stabilizations. In their model, two groups must agree on how to share the cost of fiscal adjustments; stabilization is delayed until one group concedes and bears a disproportionate share of the burden. Our work differs along three dimensions. First, in their model stabilization is needed because an exogenous shock lowers tax revenues. In our paper, debt unsustainability is an endogenous outcome. Second, in their model, the burden of the stabilization is shared according to fixed proportions, while we assume that cuts are proportional and depend on past expenditures. Third, in their model, political gridlock leads to inaction. In our model, the dilution channel implies that the incumbent delays stabilization by spending even more (see Section VIB).

## II. Fiscal Consolidations

Governments facing an unsustainable fiscal trajectory must, sooner or later, go through fiscal consolidation. Adjustments can be triggered either automatically, such as through sequestration, or by a deliberate government decision. By providing descriptive evidence, we show that in both cases, spending cuts tend to be proportional, thus preserving the priorities established in previous budgets. These findings will serve as a basis for our modeling assumptions.

[11] Bowen, Chen, and Eraslan (2014) study legislative bargaining over public spending; they find that the public good is efficiently provided in the long run when its status quo level is endogenous. Azzimonti, Karpuska, and Mihalache (2023) assume a fixed status quo for the public good and an endogenous status quo level for tax rates and transfers. They find that politicians "overuse" the inertial policy instruments to constrain future governments.

### A. *Automatic Mechanisms: US Sequestration*

Governments use budget enforcement mechanisms to control spending and deficits. In the United States, so-called sequestration causes automatic cancellation of previously enacted spending, making largely across-the-board reductions to nonexempt programs.[12]

The US sequestration process was first introduced in 1985 by the Gramm-Rudman-Hollings Balanced Budget Act, which established automatic percentage cuts in case the budget exceeded the specified deficit target for that year (Lynch 2011). After Gramm-Rudman was discontinued in 1990, the Budget Control Act of 2011 (BCA) reinstated sequestration to reduce the deficit by at least $1.2 trillion over 10 years. The BCA established a procedure known as Joint Committee sequestration, which automatically reduced both discretionary and mandatory spending annually between 2012 and 2021 (Lynch 2013). In 2021, Joint Committee sequestration was extended to 2031 by the Infrastructure Investment and Jobs Act.

The sequestration process is regulated by strict rules, with several programs being exempt from sequestration, including Social Security, Medicaid, veterans' programs, and food stamps (sequestration for Medicare is limited to no more than 2 percent).[13] When sequestration occurred, the required cuts ($109.3 billion each year from 2014 to 2021) were evenly divided between defense and nondefense programs (see Konigsberg 2019). Within these two categories, nonexempt programs were reduced by a uniform percentage to achieve the total necessary reduction for each category ($54.67 billion). Since 2013, sequestration of nonexempt mandatory spending has been implemented every year. Table A-1 in Konigsberg (2019) shows the percentage cuts by fiscal year for defense and nondefense programs.

It is important to note that by imposing uniform spending cuts, the US sequester formula tends to preserve, in relative terms, the status quo established by previous administrations. On this point, Senator Phil Gramm, the architect of the sequester, stated, "What [the Gramm-Rudman-Hollings Act] does is simply make [...] the President the instrument of the will of Congress in sequestering across the board proportionally so as to preserve the congressional intent in term of priorities" (US Senate, October, 3, 1985; see also Stith 1988). These considerations were recognized in 2011 by Republicans, who expressed concerns that the sequestration process could benefit Democrats, who controlled a unified government from 2009 to 2011. For instance, Representative Tom Cole (R-OK) argued that "across-the-board cuts essentially mean you lose the ability to [...] prioritize. In fact, you adopt the priorities of the people that wrote the original budget. [...] That means we are adopting [Democrats'] priorities when we cut in this manner."[14]

Finally, we mention that besides Joint Committee sequestration, in the United States there are two other automatic mechanisms, known as "cap sequestration" and

[12] In Europe, several countries have adopted automatic correction mechanisms under the 2012 Fiscal Compact. The design varies across European countries; some prescribe specific adjustments after fiscal rule violations (e.g., in the Slovak Republic expenses must be reduced by 3 percent at a debt-to-GDP ratio of 55 percent), while others require the government to present corrective plans, such as in Germany and Ireland.

[13] The full list of exemptions is found in Sections 255 and 256 of the Gramm-Rudman-Hollings Act. Also, automatic tax increases are not included in the sequestration process.

[14] 157 *Congressional Records* H1228 (2001).

"PAYGO sequestration." Cap sequestration was introduced by the BCA in 2011, which set discretionary spending limits for each year from 2012 to 2021. If Congress appropriated more funds than permitted, sequestration would eliminate the excess. The second mechanism was introduced by the Statutory Pay-As-You-Go Act of 2010, which mandates that any new mandatory spending increases or tax reductions must be entirely offset by other measures, ensuring that the overall effect is budget neutral. Failure to meet this requirement results in the sequestration of nonexempt mandatory programs. Unlike Joint Committee sequestration, the enforcement of cap and PAYGO sequestrations has been regularly waived by Congress. This was the case, for example, for both the $1.9 trillion American Rescue Plan Act of 2021 and the 2017 Trump tax cuts.

### B. *Deliberate Consolidations*

We now examine whether fiscal cuts resulting from deliberate consolidations also follow a proportional pattern. First, we rely on Alesina, Favero, and Giavazzi (2019) to identify the instances of fiscal consolidations (for more details, see online Appendix I). By analyzing narrative records (e.g., executive speeches and parliamentary reports), Alesina, Favero, and Giavazzi (2019) built a dataset of fiscal consolidation episodes from 1978 to 2014 across 16 OECD countries. Since austerity plans are almost always multiyear, based on Alesina, Favero, and Giavazzi (2019), we determine each consolidation's start year and end year. Consolidation episodes are listed in Table 1 in online Appendix I. While Alesina, Favero, and Giavazzi (2019) exploited aggregate spending data, we use the COFOG (Classification of the Functions of Government) data, developed by the OECD, which splits expenditure data into ten "functional" groups.[15]

To examine the extent to which functions are reduced during consolidations, Figure 1 displays the function share over total spending in the year before the start of the consolidation (x-axis) and in the final year of the consolidation (y-axis). Due to the limited availability of disaggregated COFOG data, we have a total of 25 consolidation episodes. For most consolidations, the observations lie close to the 45-degree line, suggesting that the functions' shares remain roughly constant after a consolidation. This means that cuts tend to be across the board and proportional, and consequently, consolidations broadly preserve the ranking of spending programs in terms of size. We observe that, if anything, larger programs (mainly social protection, which includes Social Security programs) are cut relatively less.

Finally, we investigate whether the ideology of the party carrying out a consolidation affects the type of functions that are cut. For this analysis, the number of episodes is further reduced since we exclude fiscal consolidations that underwent political majority changes during the consolidation period. Based on party-issue ownership, we assign each of the ten spending functions to either a left-wing, right-wing, or neutral ideology. Following Table 4A in Epp, Lovett, and Baumgartner (2014b),

[15]The functions are general public services, defense, public order and safety, economic affairs, environmental protection, housing and community amenities, health, education, social protection, and recreation, culture, and religion. For more information about the types of expenditures included in each function, see Annex C of OECD (2021).

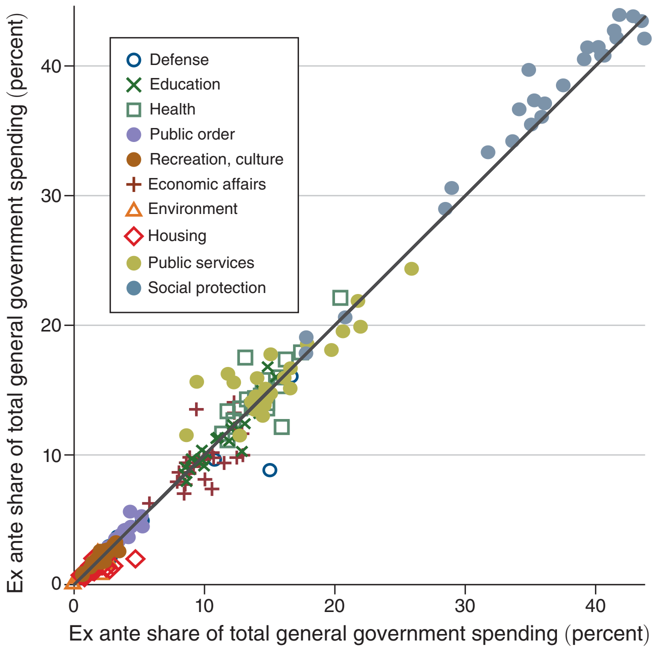


Figure 1. Fiscal Consolidations: Spending Shares

we classify defense and public order as right-wing priorities, and we classify environment, housing, health, education, and social protection as left-wing priorities. With the caveat that we have a limited number of consolidation episodes, in online Appendix I, we argue that the ideology of the party responsible for the consolidation does not seem to play a major role; we find that right-wing parties do not cut more left-wing programs (and vice versa).

## III. Spending Persistence

The canonical model of Tabellini and Alesina (1990) assumes no spending stickiness and consequently implies significant shifts in spending patterns following government changes. In this paper, by contrast, we assume that spending is inertial. To support this assumption, we examine the persistence of spending in the US budget based on data from the Office of Management and Budget (OMB) from 1977 to 2022. The OMB categorizes US federal spending into 20 functions, where each function consists of spending programs serving common purposes or activities, such as health and national defense. To measure spending persistence, we calculate the one-year lag autocorrelation for each budget function's share of the total budget authority. Based on party-issue ownership, and following Epp, Lovett, and Baumgartner (2014b), we assign each function to either the Republican or Democratic parties or a neutral category. Column 3 of Table 2 in online Appendix J shows that four functions have a one-year lag autocorrelation above 0.9. These functions together account for about 40 percent of the budget. Two of the most inertial functions are "owned" by the Republican Party, namely, National Defense and Administration of Justice (which includes programs such as US Customs and Border Protection), while the other two functions are owned by the Democratic Party, namely Health (including Medicaid) and Medicare. More generally, looking at Table 2 in online Appendix J, autocorrelation is positive and substantial for most other functions, providing evidence of budgetary inertia.[16]

It is interesting to add that, according to Jones, True, and Baumgartner (1997), the US budgeting process has become more inertial over time. By examining more disaggregated spending data (namely, 55 nondefense OMB subfunctions), they argue that this result cannot be attributed to institutional gridlock resulting from more frequently divided governments. Instead, they suggest that one contributing factor is the increasing proportion of mandatory spending.

The finding that most functions are highly persistent explains why we observe limited shifts in spending patterns following government changes, despite political partisanship. To investigate the existence (or lack thereof) of partisan budgeting, several papers have regressed spending for various budget functions on measures of government ideology (e.g., dummy variables for left-wing governments). The survey article by Potrafke (2017) concludes that there is weak evidence of partisan budgeting, particularly in recent decades.[17] Along similar lines, Epp, Lovett, and Baumgartner (2014a) analyze spending changes across subfunctions of the US budget and find many of them to be "inconsistent" based on the governing party's ideology, even during periods of unified governments. In their views, this suggests that government spending is influenced not only by partisan choices but also by formulas that maintain or increase spending on certain programs when more people become legally entitled to them.

## IV. The Model

We consider a two-period model: $t = 1, 2$. In the first period, public spending is financed by current taxes and debt. In the second period, any remaining debt must be paid. The two-period assumption is stark, but it captures the two phases that would be observed in the infinite horizon: an early phase when the government runs deficits and a later one in which spending must be cut. In the online Appendix, we study and numerically simulate an infinite-horizon model, demonstrating that the qualitative results of the two-period model are robust.

Public policies are decided by two parties (denoted by $I$ and $O$) that stochastically alternate in power. We assume that party $I$ is the incumbent government in the first period. The opposition at time 1 is party $O$. At the end of the first period, there is an election, and party $I$ stays in power with exogenous probability $q$. With a complementary probability $(1 - q)$, party $O$ seizes power, and $I$ becomes the

[16] Relatedly, the work of Wildavsky (1964) has inspired an old empirical literature showing that the US congressional budgeting process is inertial and characterized by "small" incremental changes (see, for instance, Davis, Dempster, and Wildavsky 1966 and Dempster and Wildavsky 1979).

[17] The results of the empirical literature on partisan budgeting are mixed. For instance, Bove, Efthyvoulou, and Navas (2017) find that in OECD countries, left-wing governments allocate more resources to old-age pensions, family, and disability benefits. On the other hand, Lambertini (2003), both in the United States and in the OECD overall, finds no evidence that more liberal governments spend more on Social Security and welfare or that more conservative governments spend more on defense. After reviewing 100 panel studies, Potrafke (2017) concludes that the effect of government ideology on the type of expenditure is not significant in many of these studies.

opposition party. Throughout most of the paper, we assume that the two parties have opposite preferences over the desired composition of public spending. There are two types of goods, and each party would like to allocate the whole budget to one of them. For example, the two goods could represent partisan expenditures that target each party's constituency. In American politics, the traditional division is between Democrats, who favor spending on Social Security, health care, and education, and Republicans, who favor defense spending, border protection, and tax cuts (Pew Research Center 2019). For most practical purposes, in our setting tax breaks can be considered equivalent to spending; "tax expenditures" and tax cuts favor specific constituencies, raise debt, and cannot be easily undone because they run through the tax code (see Tanzi 2008 and OECD 2010).

We denote by $g^I$ and $g^O$ the good that is valued by party $I$ and party $O$, respectively. The per period utility of the two parties is

$$u_j(g^j) = \frac{(g^j)^{1-\sigma} - 1}{1-\sigma} \tag{1}$$

with $j = I, O$ and $\sigma \in (0,1)$. The case where parties are less than fully polarized, where each party partly enjoys the good preferred by the other party, will be discussed in Section VC. Each party discounts the future using the factor $\beta = 1$.

We assume that the tax revenue is exogenous and equal to $\tau$ in both periods. The assumption that taxes are exogenous is not important for the main point of our paper, but it greatly simplifies the analysis. To avoid cluttered notation, we also assume that the interest rate and initial debt are both zero. The government budget constraints at $t = 1$ and $t = 2$, respectively, are

$$g_1^I + g_1^O \leq \tau + b_1 \tag{2}$$

and

$$g_2^I + g_2^O \leq \tau - b_1, \tag{3}$$

where $b_1$ is the stock of debt issued at $t = 1$. To ensure solvency, assume $b_1 \leq \tau$ so that it is always feasible to pay the outstanding debt in the final period.

Our first key departure from the literature is that, whenever fiscally sustainable, the incumbent must maintain a proportion $\alpha \in [0, \bar{\alpha}]$ of the previous year's spending programs:

$$g_t^j \geq \alpha g_{t-1}^j \tag{4}$$

with $j = I, O$ and $t = 1, 2$. The parameter $\alpha$ summarizes the extent of budgetary stickiness and captures, in a reduced form, the political and legal costs of cutting existing programs. If $\alpha = 1$, budgets are highly rigid, meaning that the incumbent cannot cut the existing spending programs. When $\alpha = 0$, we obtain the canonical model of Tabellini and Alesina (1990). When $\alpha > 1$, spending programs expand over time; for example, Medicare spending might increase because the elderly population is growing. We assume here that all spending is equally sticky. We will later relax this assumption by assuming that some spending is stickier than others, potentially capturing the distinction between mandatory and discretionary spending.

Our second departure, which is required by the existence of persistence, is the explicit consideration of budget sustainability. For each period, we introduce a *sustainability constraint*, determining whether it is feasible to maintain the prior government's commitments. At $t = 2$, if

$$\alpha g_1^I + \alpha g_1^O > \tau - b_1, \tag{5}$$

the total commitments inherited from the previous period are not sustainable. Consequently, we must impose fiscal consolidation.[18]

In Section II, we have presented evidence that during consolidation (whether deliberate or automatic), cuts tend to be proportional and applied across the board. Consistent with this, we assume that during consolidation (or sequestration, terms we use interchangeably), each program is reduced by the same proportion $\nu \in [0,1)$:

$$g_2^I = \alpha g_1^I \underbrace{\left(\frac{\tau - b_1}{\alpha g_1^I + \alpha g_1^O}\right)}_{\equiv \nu} \quad \text{and} \quad g_2^O = \alpha g_1^O \underbrace{\left(\frac{\tau - b_1}{\alpha g_1^I + \alpha g_1^O}\right)}_{\equiv \nu}, \tag{6}$$

where $\nu$ is endogenous to the model and indicates the severity of the needed consolidation: the lower $\nu$, the harsher the consolidation. Notice that upon consolidation, the incumbent government has no fiscal space because all available resources have already been committed, leaving no room for discretionary choices regarding which programs to curtail. This assumption is supported by the descriptive evidence discussed in Section IIB and is coherent with the sequestration rule in US budgeting, which prescribes that the president cannot alter the spending priorities previously established by Congress.[19] In the online Appendix, we modify equation (6) to incorporate some of the complexities of the sequestration formula discussed in Section IIA. As we will discuss in Section VI, the results remain qualitatively unchanged, and in fact, they may be strengthened.

One can rewrite (6) as

$$g_2^I = (\tau - b_1) \underbrace{\left(\frac{g_1^I}{g_1^I + g_1^O}\right)}_{\equiv \psi} \quad \text{and} \quad g_2^O = (\tau - b_1) \underbrace{\left(\frac{g_1^O}{g_1^I + g_1^O}\right)}_{\equiv 1-\psi}. \tag{7}$$

The shares $\psi$ and $1 - \psi$ are endogenous to the model, but they are predetermined at time $t = 2$. The key observation from (7) is that upon consolidation, each spending program receives a share of available resources in proportion to its relative size. The

[18] In online Appendix D, we allow the incumbent to avoid cutting entitlements by running more debt. In the online Appendix, we also examine a model where it is possible to default on debt but there is a cost involved, and the incumbent can choose between cutting entitlements or defaulting on debt. We show that the thrust of our results remains unchanged.

[19] If the incumbent government were able to cut spending upon sequestration selectively, we would introduce an artificial discontinuity: when (5) holds with equality, the incumbent would have no fiscal space, but it would be able to allocate spending freely as soon as that inequality becomes strict. The thrust of our results will be maintained if we assume that the ability of the incumbent to cut spending selectively is less than full.

exact specification of equation (7) is not essential for many of our results. What is required, more generally, is that the share $\psi$ increases with $g_1^I$.[20]

At $t = 1$, the sustainability constraint takes into account that future entitlements constitute a liability for the government. Let $g_0^O$ and $g_0^I$ be the spending levels that were chosen in (unmodeled) period 0. We assume that these entitlements are not sustainable if the present value of "committed" spending is higher than the present value of taxes over the two periods:

$$(8) \qquad (\alpha + \alpha^2) g_0^I + (\alpha + \alpha^2) g_0^O > 2\tau.$$

To understand this expression, recall that the interest rate is zero and that by (4) entitlements depreciate (when $\alpha < 1$) or grow (when $\alpha > 1$) geometrically. When (8) holds, past commitments are not sustainable in the long run, and spending programs must be cut to restore sustainability. (In online Appendix E, we define the sustainability constraint for the infinite-horizon model.) Notice that there is a key difference between (5) and (8) regarding the urgency of sequestration. In the final period, a sequestration cannot be postponed. If instead (8) holds, politicians might be tempted to ignore the constraint, run debt to pay current obligations, and postpone the sequestration. We consider this possibility of delaying sequestration in online Appendix D (two-period model) and in online Appendix E (for the infinite-horizon model). In the main text, we abstract from these considerations by assuming that obligations from $t = 0$ are sustainable.

ASSUMPTION 1: *$g_0^I$ and $g_0^O$ are sufficiently small so that equation* (8) *does not hold for all $\alpha \in [0, \bar{\alpha}]$.*

Figure 2 summarizes the timeline. Given $g_0^I$ and $g_0^O$, at $t = 1$, party *I* selects $g_1^I$ and $g_1^O$ subject to (2) and (4). At $t = 2$, party *I* stays in power with probability $q$, and party *O* becomes the new incumbent with probability $1 - q$. If (5) holds, a sequestration occurs. If (5) does not hold, the party in office chooses policies $g_2^I$ and $g_2^O$ subject to (3) and (4).

*Solution.*—As a benchmark, we start by computing the optimal allocation. We solve the problem of a social planner who maximizes the weighted sum of both parties' utilities. Assuming a discount rate equal to the interest rate ($\beta = 1$ and a zero interest rate), and constant tax revenue, it is easy to show that a social planner would choose zero debt. If both parties are equally weighted by the planner, the optimal allocation is $g_t^j = \tau/2$ for $t = 1, 2$ and $j = I, O$. More generally, for arbitrary weights in the planner's problem, the ratio $g_t^I / g_t^0$ must be constant over time.

We now solve the politico-economic model backward, starting from the final period. At $t = 2$, from (2) and (5), we obtain that a sequestration occurs if

$$(9) \qquad b_1 > \tilde{b} \equiv \frac{\tau(1 - \alpha)}{(1 + \alpha)},$$

[20] For deliberate consolidations, this could be rationalized by supposing that the two parties bargain over the share $\psi$. A party with larger spending commitments would have a stronger outside option and appropriate a larger share of the available resources. We thank a referee for suggesting this interpretation.

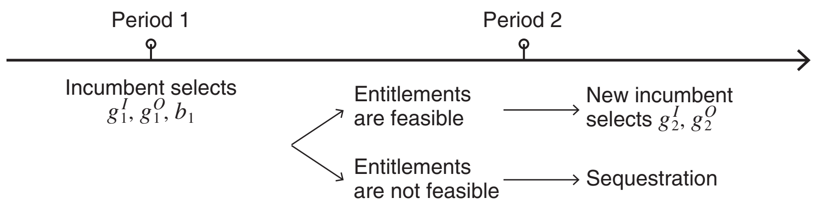


FIGURE 2. TIMELINE

where $\tilde{b}$ is the sequestration threshold. When $b_1 \in (\tilde{b}, \tau]$, previous governments' commitments are not sustainable and must be cut. The threshold is decreasing in $\alpha$. This is intuitive; when $\alpha$ is small, sequestration occurs only for very high levels of past spending. Define the indicator function $\Phi$, which takes the value one when a sequestration occurs at $t = 2$ and the value zero otherwise.

At $t = 2$, available resources are $\tau - b_1$. It is straightforward to compute the consumption of party $I$ in the final period. If there is no sequestration and $I$ stays in power, she will have to (partially) maintain the opposition's entitlements and keep the remaining resources for herself; then $g_2^I = \tau - b_1 - \alpha g_1^O$. If, instead, $I$ loses power, $g_2^I = \alpha g_1^I$. Finally, when there is sequestration, spending programs are downsized using the sequestration rule in (7). Summarizing,

$$g_2^I = \begin{cases} \tau - b_1 - \alpha g_1^O, & \text{if } \Phi = 0 \text{ and } I \textit{ stays} \text{ in power} \\ \alpha g_1^I, & \text{if } \Phi = 0 \text{ and } I \textit{ loses} \text{ power} \\ \psi(\tau - b_1), & \text{if } \Phi = 1 \end{cases}.$$

Moving to the first period, party $I$'s dynamic problem can be written as

$$(10) \quad \max_{\{g_1^O, g_1^I, b_1\}} \Big\{ u(g_1^I) + (1 - \Phi)\big[qu(\tau - b_1 - \alpha g_1^O) + (1 - q)u(\alpha g_1^I)\big] + \Phi u(\psi(\tau - b_1)) \Big\}$$

subject to (2), (4), and $b_1 \leq \tau$.

There are two areas to consider: when $b_1$ is below the sequestration threshold and when it is above. When $b_1 \leq \tilde{b}$, so that $\Phi = 0$, after noticing that party $I$ optimally allocates $\alpha g_0^O$ to the opposition, the first-order condition equalizing the marginal benefit and cost of debt is

$$(11) \quad (\tau + b_1 - \alpha g_0^O)^{-\sigma} + (1 - q)\alpha\big[\alpha(\tau + b_1 - \alpha g_0^O)\big]^{-\sigma} = q(\tau - b_1 - \alpha^2 g_0^O)^{-\sigma}.$$

The first term on the left-hand side of (11) indicates that higher $\alpha$ raises the marginal utility of consumption today. If the incumbent wants to achieve a given target for current consumption, she needs to raise more debt. The second term on the left-hand side reflects the "insurance" effect: higher spending today secures higher spending in the next period in case party $I$ is voted out of office. The right-hand side of (11) is the future marginal cost of debt. As emphasized by the strategic models of debt, a higher probability of losing power (lower $q$) makes the incumbent more myopic, raising the deficit bias.

When $b_1 > \tilde{b}$, so that $\Phi = 1$, the first-order condition is given by

$$(12)\qquad \left(\tau + b_1 - \alpha g_0^O\right)^{-\sigma} = \left[\psi(\tau - b_1)\right]^{-\sigma} \underbrace{\left[\psi - \overbrace{\frac{\alpha g_0^O}{(\tau + b_1)^2}}^{=\partial\psi/\partial g_1^I \geq 0}(\tau - b_1)\right]}_{dilution}.$$

As in (11), the first term on the left-hand side of (12) is the marginal utility of consumption today. Because political risk is eliminated upon sequestration, the right-hand side does not include $q$ anymore. Regardless of the electoral outcome, party $I$ obtains a proportion $\psi = g_1^I/(g_1^I + g_1^O)$ of available resources. From (12), we obtain that upon sequestration the (endogenous) slope of good $I$ over time is given by

$$(13)\qquad \frac{g_2^I}{g_1^I} = \left[\psi - \frac{\partial\psi}{\partial g_1^I}(\tau - b_1)\right]^{1/\sigma}.$$

The presence of the opposition's entitlements makes party $I$ more myopic, resulting in a downward tilt of the above slope. To illustrate this effect, consider the following two cases: when $g_0^O = 0$ and when $g_0^O > 0$. When the opposition has no entitlements, upon sequestration, all available resources will be allocated to $I$'s preferred good. Because $\psi = 1$ and there is no dilution, the right-hand side of (13) becomes one. In this case, the incumbent fully internalizes the cost of debt and wants to perfectly smooth spending over time. When instead $g_0^O > 0$, and hence, $\psi < 1$, the incumbent no longer internalizes the whole cost of debt because saved resources will partly benefit the opposition. Moreover, there is a dilution effect; because $\partial\psi/\partial g_1^I > 0$, more spending today dilutes the opposition's spending share upon sequestration. Due to both channels, the opposition's entitlements lower the marginal cost of debt and decrease the right-hand side of (12) and (13), giving incentives to front-load spending.

Conditions (11) and (12) characterize the solution when equilibrium debt is, respectively, below and above the sequestration threshold. In some cases, however, the solution is not characterized by either (11) or (12). This occurs when (9) is satisfied with equality, and optimal debt exactly coincides with the sequestration threshold, a point where the objective is not differentiable.

From the first-order conditions above, we examine how inertia affects politicians' strategic incentives to spend, specifically through the dilution and insurance channels. First, because the opposition's entitlements at $t = 2$ increase with $\alpha$, budgetary inertia affects the dilution channel. The effect of $\alpha$ on the insurance channel is ambiguous because there are, roughly speaking, substitution and income effects pushing in opposite directions. On one hand, looking at the second term on the left-hand side of (11), lower stickiness makes spending less effective in constraining future governments, inducing less spending and debt. On the other hand, with a lower $\alpha$, a given target level of future spending is achieved with more debt, which increases the incentives to overspend to lock in the subsequent government and reduce political risk.

A widely held view is that reducing budget inertia would lead to smaller debt levels. Taking all effects into account, we find that this may not necessarily be the case.

PROPOSITION 1 (Benchmark characterization):

(*i*) *Equilibrium debt when* $\alpha = 0$ *is lower than debt when* $\alpha = 1$ *if the preexisting opposition's entitlements are sufficiently large:*

$$g_0^O > \frac{2\tau(1-q)}{\left(1+q^{\frac{1}{\sigma}}\right)^2}. \tag{14}$$

(*ii*) *Debt is increasing in* $\alpha$ *if stickiness is sufficiently close to zero. Debt is a nonmonotone function of* $\alpha$.

PROOF:

See Appendix A.

The above proposition provides a simple condition under which stickiness exacerbates the debt problem: when political turnover is low and the opposition's entitlements are high. This result arises because a higher $q$ makes the incumbent less myopic when $\alpha = 0$, while a higher $g_0^O$ makes the incumbent more myopic when $\alpha = 1$ and sequestration occurs.

Figure 3 illustrates the relation between stickiness (x-axis) and debt (y-axis).[21] The dashed red line is the sequestration threshold, which from (9) is decreasing in $\alpha$. In panel A, we assume that $g_0^O = 0$ and thus abstract from the dilution channel. The insurance channel implies a hump-shaped relation between $\alpha$ and debt. When $\alpha$ is low, triggering a sequestration would require implausibly high spending levels. In this range, the relevant first-order condition is (11), and we find that—under the assumption made throughout the paper that $0 < \sigma < 1$—equilibrium debt increases with $\alpha$; higher stickiness leads to more debt accumulation.[22]

As stickiness increases, however, the sequestration threshold declines, and it becomes optimal to trigger a sequestration. The incumbent eliminates political risk by leaving no fiscal space to the successor and choosing a debt level exactly equal to the sequestration debt threshold. Since this threshold is decreasing in $\alpha$, an increase in $\alpha$ makes it "cheaper" to lock in subsequent governments and fully eliminate political risk, resulting in reduced spending incentives. When $\alpha \geq 1$, stickiness gives party *I* commitment and allows it to implement the first-best solution: as a result, debt is zero.[23]

[21] We set $g_0^O = 0$ (panel A) and $g_0^O = 0.5$ (panel B), $q = \sigma = 1/2$, and $\tau = 1$.

[22] Tabellini and Alesina (1990, p. 43) also assume that $\sigma \in (0,1)$. In online Appendix L, we show that when $\sigma > 1$, we achieve results that are comparable to Figure 3, with one main difference: for low values of $\alpha$, debt decreases as stickiness rises.

[23] While efficient, this allocation strongly favors party *I*. In Section VA we will show that a supermajority requirement leads to a more equitable allocation on the Pareto frontier.

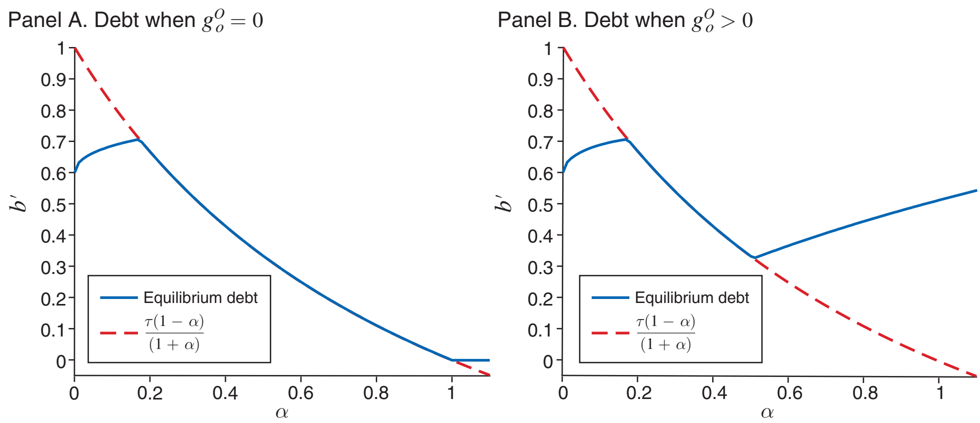


Figure 3. Debt Accumulation

In panel B of Figure 3, we assume that $g_0^O > 0$. We show that for large $\alpha$ a new effect dominates the other channel; as $\alpha$ increases, the opposition's entitlements (a combination of $g_0^O$ and $\alpha$) become more important, giving the incumbent the incentive to raise debt above the sequestration threshold to dilute the opposition's entitlements.

Note the different roles played by the sequestration when there are no preexisting commitments (panel A) versus when they are positive (panel B). In panel A, when $\alpha \leq 1$, cuts never happen in equilibrium. In panel B, sequestration is actively used on the equilibrium path to dilute past entitlements and, consequently, further fuels the incentives to overaccumulate debt.

The dilution motive is a novel channel not present in the related literature. In the canonical strategic model of debt with $\alpha = 0$, the incumbent is constrained only by the inherited stock of debt; how spending was allocated by past governments is completely irrelevant. In our model, with initial debt fixed, a higher $g_0^O$ leads to a stronger incentive for debt accumulation. The dilution channel might imply a spending spiral in which higher opposition entitlements lead to overspending and more debt. To see this more clearly, using (12), we provide a simple closed-form solution for debt above the sequestration threshold when $\sigma \rightarrow 1$:

$$b_1^* = \frac{-\tau + \sqrt{\tau^2 + 4\tau\alpha g_0^O}}{2}, \tag{15}$$

which shows that debt is increasing in $g_0^O$ when $\alpha$ is close to one.[24] In other words, with a looming fiscal stabilization, a loose fiscal policy leads to even looser policies by subsequent governments. One could view Obama's and Trump's fiscal policies through this lens; i.e., the surge in Social Security and health program entitlements under the Obama administration was followed by the Trump tax cuts and other spending measures that further bloated US debt. In Section VIC, we will present

[24] If initial debt $b_0$ were positive, the dilution motive would still be present; expression (15) would be decreasing in $b_0$ and increasing in $g_0^O$.

anecdotal and descriptive evidence consistent with the strategic mechanisms outlined in our theory.

The main assumption underlying the dilution effect is that a political party can improve its position in a consolidation by increasing the spending on its preferred programs before the consolidation takes place. Interestingly, in 2011, Senators Bob Corker and Claire McCaskill sponsored the Commitment to American Prosperity Act (S245), proposing an alternative sequestration formula whereby cuts to each spending category should be proportionate to the rate of growth in each category from the previous year. Had it been approved, this would have penalized programs that had experienced faster growth, weakening the incentives to boost spending before sequestration.

## V. Extensions

In this section, we extend the basic model along three dimensions. In Section VA, we introduce a supermajority requirement for increasing spending. In Section VB, we allow politicians to avoid sequestration by cutting less inertial spending. In Section VC we study a model with less than full preference polarization.

### A. *Checks and Balances*

In the previous analysis, we assumed that spending is downwardly rigid but the party in office can unilaterally increase spending. In practice, however, executives may need the opposition's approval to establish new spending programs or to expand the current ones. For example, in the United States, it is common to have divided governments, implying that both parties must approve a policy change. To analyze this issue, we modify the decision process. We introduce a supermajority requirement for increasing spending: the executive has the power to propose a policy change, but any such change must also be approved by the opposition. The timing of events is as follows:

(i) *Period* $t = 1$. The executive (party $I$) makes a take-it-or-leave-it offer $(g_1^O, g_1^I, b_1)$ to party $O$, which must satisfy $g_1^O \geq \alpha g_0^O$ and $g_1^I \geq \alpha g_0^I$. If party $I$'s proposal is rejected, spending is equal to $\left(\alpha g_0^I, \alpha g_0^O\right)$, and debt is residually determined by the budget constraint (2).

(ii) *Period* $t = 2$. Party $I$ stays in power with probability $q$, and with probability $1 - q$, party $O$ becomes the new incumbent government. Whenever (5) holds, past entitlements are not sustainable, and there is sequestration as discussed in (6). If past entitlements are instead sustainable, the incumbent makes a proposal $(g_2^O, g_2^I)$ that satisfies (3) and (4). If the proposal is rejected, spending is equal to $(\alpha g_1^I, \alpha g_1^O)$.

We solve the model backward. In the second period, the bargaining outcome is straightforward. Suppose, for instance, that party $I$ is still in power at $t = 2$. If there is no sequestration, party $I$ gives the opposition her outside option $(\alpha g_1^O)$ and keeps the remaining resources for herself.

In the first period, the incumbent chooses $\left(g_1^O, g_1^I, b_1\right)$ by solving problem (10) with the additional constraint that her proposal must be accepted:

$$\text{(AC)} \qquad u\left(g_1^O\right) + (1 - \Phi)\left[qu\left(\alpha g_1^O\right) + (1 - q)u\left(\tau - b_1 - \alpha g_1^I\right)\right] + \Phi u\left(\left(\tau - b_1\right)(1 - \psi)\right) \geq u\left(\alpha g_0^O\right) + \left[qu\left(\alpha^2 g_0^O\right) + (1 - q)u\left(\tau - b_1^r - \alpha^2 g_0^I\right)\right].$$

The acceptance constraint (AC) implies that the opposition accepts the incumbent's proposal if the utility of accepting is greater than or equal to the outside option. If negotiations break down, the opposition obtains $\alpha g_0^O$ and moves to the next period with a level of debt equal to $b_1^r$, where $b_1^r = \alpha g_0^O + \alpha g_0^I - \tau$. The next proposition states a stark result: budgetary stickiness, together with checks and balances, eliminates the deficit bias.

PROPOSITION 2: *Assume* $\alpha = 1$. *When the incumbent needs the opposition's approval to pass a policy change, equilibrium debt is zero for any initial entitlements* $(g_0^O, g_0^I)$, *spending remains constant over time, and both parties are insured against political risk.*

PROOF:
See Appendix B.

Figure 4 illustrates debt accumulation as a function of $\alpha$ in the model under supermajority rule ("Bargaining solution") and in the baseline model solution.[25] When $\alpha = 1$ and $g_0^O > 0$, the two models yield distinct implications regarding debt. In the baseline model without bargaining, we showed that party *I* selects an inefficient allocation for dilution purposes. With a supermajority requirement, instead, efficiency is obtained: debt is exactly zero, and the parties' ratio of marginal utilities across the two periods is equalized. The dilution motive is not present anymore because party *I* cannot obtain the opposition's approval to increase $g_1^I$. Note that the level of $g_0^O$ has different implications in the two models. In the absence of a supermajority requirement, a higher $g_0^O$ would lead to more inefficiencies due to the dilution effect. Conversely, when a supermajority is needed, a higher $g_0^O$ would preserve the zero-debt result and, by raising the opposition's bargaining power, would lead to a more equitable spending allocation between the two parties. When $\alpha$ is smaller than one, Figure 4 illustrates that debt is still positive but lower than without bargaining.[26] Debt is lower because the acceptance constraint makes it more costly for party *I* to accumulate debt: party *O* agrees to increase debt only if she is given higher $g_1^O$.

[25] We set $g_0^O = 0.3$, $g_0^I = 0.1$, $q = \sigma = 1/2$, and $\tau = 1$.

[26] Piguillem and Riboni (2021) also find that deficit bias is reduced when the party in office negotiates policies with the opposition. In that paper, we abstract from budgetary inertia, assuming that past spending can be cut at no cost, and focus on a very different question: how fiscal rules affect debt accumulation.

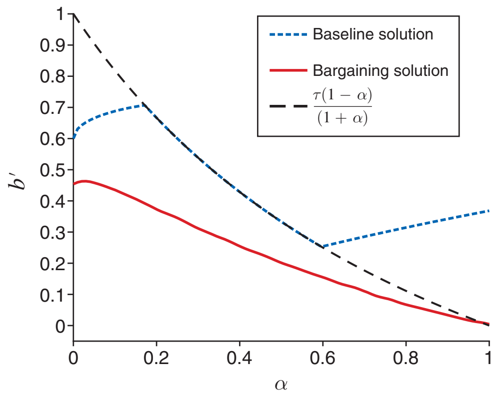


Figure 4. Bargaining versus No Bargaining

## B. *Mandatory and Discretionary Spending*

In the previous analysis, we assumed that all spending is equally sticky. We now relax this assumption to model the distinction between discretionary and mandatory spending. Since discretionary programs require annual appropriation bills, they are often assumed to be less inertial than entitlements, which are mandated by law and continue from year to year until a new law is passed that alters them.

We consider an economy with four goods. Two of them, $d^I \geq 0$ and $d^O \geq 0$, are nonsticky and can be cut at no cost by the current government. As before, the remaining goods, $g^O$ and $g^I$, are sticky, meaning that the government must maintain a fraction $\alpha$ of past spending. The thrust of the analysis would be maintained if we assume that spending for $d^I$ and $d^O$ is also sticky but to a lesser degree than for goods $g^I$ and $g^O$.

There is full polarization: party $I$ (party $O$) only values goods $g^I$ and $d^I$ (goods $g^O$ and $d^O$). As in the baseline model, assume that the incumbent does not need to negotiate policies. This implies that the party in power will not provide the nonsticky good valued by the opposition. For party $j = I, O$, the per period utility is given by

$$u_j(g^j, d^j) = \frac{(g^j)^{1-\sigma} - 1}{1-\sigma} + \theta \frac{(d^j)^{1-\sigma} - 1}{1-\sigma}, \tag{16}$$

where $\theta > 0$ is the relative utility weight attached to noninertial programs. As discussed above, we can interpret $d^I$ and $d^O$ as discretionary spending, while goods $g^O$ and $g^I$ can now be interpreted as mandatory spending. It is important to emphasize that the equivalence between discretionary spending and nonsticky spending is more nuanced in practice. Certain discretionary programs, such as defense, can also exhibit significant stickiness in the data due to the political costs associated with reducing such programs. The budget constraints are

$$g_1^O + g_1^I + d_1^I + d_1^O \leq \tau + b_1 \tag{17}$$

$$g_2^O + g_2^I + d_2^I + d_2^O \leq \tau - b_1. \tag{18}$$

We assume that mandatory spending is prioritized over discretionary spending so that sequestration is triggered only if, after reducing discretionary spending to zero, remaining resources are not sufficient to pay existing entitlements.[27] As before, at $t = 2$, a sequestration occurs if $\alpha\left(g_1^O + g_1^I\right) \geq \tau - b_1$. Using (17) and since party $I$ optimally sets $d_1^O = 0$, a sequestration occurs if

$$b_1 \geq \frac{\tau(1-\alpha)}{(1+\alpha)} + \frac{\alpha d_1^I}{(1+\alpha)}. \tag{19}$$

Note that the sequestration threshold now depends on the (endogenous) amount of time 1 discretionary spending. The planner's solution is again zero debt and a smooth path of spending. Moreover, for $j = I, O$ and $t = 1, 2$, intratemporal efficiency requires that

$$\frac{d_t^j}{g_t^j} = \theta^{\frac{1}{\sigma}}.$$

This section shows that party $I$ will deviate from the planner's solution by selecting a strictly positive level of debt and by *overproviding mandatory spending*.

We solve the equilibrium backward. If there is a sequestration at $t = 2$, discretionary spending will be zero for both parties, and entitlements will be cut. Party $I$ will obtain a share $\psi = g_1^I/\left(g_1^I + g_1^O\right)$ of available resources. If there is no sequestration, the incumbent at $t = 2$ retains some discretion to allocate resources after satisfying past commitments. In this case, let $V_2^I\left(g_1^I, b_1\right)$ be the time 2 value function of party $I$ if she stays in power:

$$V_2^I\left(g_1^I, b_1\right) = \max_{\left\{g_2^I, d_2^I\right\}} u\left(g_2^I\right) + \theta u\left(d_2^I\right) \tag{20}$$

subject to

$$g_2^I + d_2^I + \alpha^2 g_0^O \leq \tau - b_1$$

$$g_2^I \geq \alpha g_1^I.$$

Let $\lambda$ denote the multiplier associated with the "sticky constraint" $g_2^I \geq \alpha g_1^I$, which will be binding when $\alpha$ is sufficiently large. We obtain

$$g_2^I = \max\left\{\frac{1}{1+\theta^{\frac{1}{\sigma}}}\left(\tau - b_1 - \alpha^2 g_0^O\right), \alpha g_1^I\right\} \tag{21}$$

$$d_2^I = \tau - b_1 - \alpha^2 g_0^O - g_2^I.$$

[27] We could also add a required minimum level of discretionary spending, without changing the results.

We now move to the first period.

$$\max_{\{g_1^I, d_1^I, b_1\}} \Big\{ u(g_1^I) + \theta u(d_1^I) + (1 - \Phi)\big[q V_2^I(g_1^I, b_1) + (1 - q) u(\alpha g_1^I)\big] + \Phi u\big((\tau - b_1)\psi\big) \Big\}$$

subject to

$$\tau + b_1 \geq g_1^I + d_1^I + \alpha g_0^O.$$

We begin by considering the case in which the incumbent chooses allocations in the no-sequestration region. In this section, we provide the main equations to help intuition; see online Appendix C for more details. Knowing that $\partial V_2^I / \partial g_1^I = -\alpha\lambda$, the intratemporal and intertemporal allocations are characterized by

$$\text{(22)} \qquad \left[1 + (1 - q)\alpha^{1-\sigma}\right]\left(g_1^{I*}\right)^{-\sigma} = q\alpha\lambda + \theta\left(d_1^{I*}\right)^{-\sigma}$$

$$\text{(23)} \qquad d_2^{I*} = q^{\frac{1}{\sigma}} d_1^{I*}.$$

We obtain several results. First, condition (23) implies that when $I$ stays in power (and there is no sequestration), discretionary spending decreases over time. Intuitively, a higher probability of losing power implies that the incumbent will front-load discretionary spending more. Condition (23) resembles the dynamics in the standard model a la Alesina and Tabellini (1990), with only discretionary spending. Surprisingly, equation (23) implies that the ratio does not vary with $\alpha$. As is clear from equation (22), budgetary persistence does affect the intratemporal allocation between mandatory and discretionary spending, but given this "level" distortion, the dynamics of discretionary spending is unaffected.

Second, mandatory spending is overprovided compared to efficiency. Two mechanisms are affecting the intratemporal allocation. The first one, captured by the term $(1 - q)\alpha^{1-\sigma}$ in equation (22), is the insurance effect due to the presence of turnover risk that can be hedged with mandatory spending. The second one is the term $q\alpha\lambda$ in the same equation, which captures the possibility that if the incumbent remains in power, she will not be able to provide her desired level of spending. The equilibrium spending ratio satisfies

$$\text{(24)} \qquad \frac{d_1^{I*}}{g_1^{I*}} = \begin{cases} \dfrac{\theta^{\frac{1}{\sigma}}}{\left[1 + (1 - q)\alpha^{1-\sigma}\right]^{\frac{1}{\sigma}}}, & \text{if } \lambda = 0 \\[2ex] \dfrac{\theta^{\frac{1}{\sigma}}(1 + \alpha)^{\frac{1}{\sigma}}}{\left(1 + \alpha^{1-\sigma}\right)^{\frac{1}{\sigma}}}, & \text{if } \lambda > 0 \end{cases}.$$

To interpret the above expression, recall that the optimal ratio is $\theta^{1/\sigma}$ and that $\lambda$ starts being positive when $\alpha$ is above a certain threshold, denoted by $\tilde{\alpha} > 0$. Start by considering values of $\alpha < \tilde{\alpha}$. Because $\lambda = 0$, the future level of mandatory spending remains uncertain. When $\alpha = 0$, the equilibrium ratio coincides with the

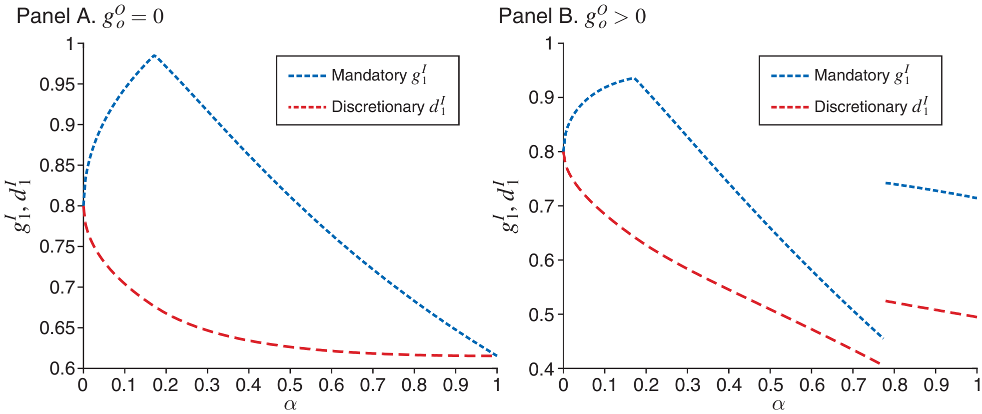


Figure 5. Time-1 Underprovision of Discretionary Spending

efficient one because mandatory spending cannot be used to smooth spending. As $\alpha$ increases, $g_1^I$ becomes more valuable as an insurance tool, so that the incumbent tilts spending toward it. When $\alpha \geq \tilde{\alpha}$, the uncertainty about $g_2^I$ disappears. Because $\lambda > 0$, independently of who will be in power, the current incumbent enjoys the same level of mandatory spending $\alpha g_1^I$ in the second period. For this reason, $q$ vanishes in the second line of (24). As long as $\alpha < 1$, the incumbent is still left with a dynamic inefficiency because to obtain one unit of consumption tomorrow, it must spend $1/\alpha$ units today. Thus, the underprovision of discretionary spending remains. But as $\alpha \rightarrow 1$, the incumbent can obtain full insurance and perfect intertemporal smoothing; the distortive effect of persistence vanishes and the intratemporal spending ratio converges to the optimal one.

The underprovision of discretionary spending is shown in Figure 5, which illustrates mandatory and discretionary spending in the first period as a function of $\alpha$. In panel A, we assume $g_0^O = 0$ and $\theta = 1$, which implies that the ratio $d_1^I/g_1^I$ should be one. When $\alpha = 0$, the ratio $d_1^I/g_1^I$ coincides with the efficient one. When $\alpha \in (0,1)$, the ratio is less than one, decreasing for small $\alpha$'s and then increasing again. In panel B, we show that when $g_0^O > 0$, there is an additional effect due to the possibility of a sequestration on the equilibrium path. We will discuss sequestration at the end of this section.

Third, as in Section IV, debt is a nonmonotone function of $\alpha$, and the dilution effect is present when $g_0^O > 0$ (see Figure 6). There are, however, some qualitative differences. To understand them, we present in (25) the closed-form solution for debt accumulation conditional on no future sequestration (see panel A of Figure 6):

$$(25) \qquad b_1^* = \begin{cases} \dfrac{\tau(1-\Xi_0) - \alpha g_0^O(\alpha - \Xi_0)}{1+\Xi_0}, & \text{if } \lambda = 0 \\ \dfrac{\tau\left(\left[1-q^{\frac{1}{\sigma}}\right]\Xi_1 + 1 - \alpha\right) - \alpha g_0^O \Xi_1\left(\alpha - q^{\frac{1}{\sigma}}\right)}{\left(1+q^{\frac{1}{\sigma}}\right)\Xi_1 + 1 + \alpha}, & \text{if } \lambda > 0 \end{cases},$$

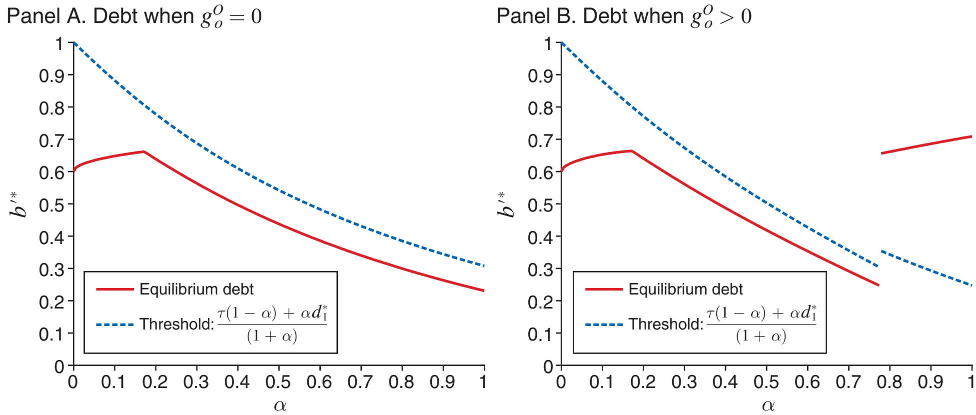


Figure 6. Debt Accumulation with Mandatory and Discretionary Spending

where,

$$\Xi_0 = \frac{q^{\frac{1}{\sigma}}\left(1 + \theta^{\frac{1}{\sigma}}\right)}{\left[1 + (1-q)\,\alpha^{1-\sigma}\right]^{\frac{1}{\sigma}} + \theta^{\frac{1}{\sigma}}} \leq 1, \quad \text{and} \quad \Xi_1 = \left[\frac{\theta(1+\alpha)}{\left(1+\alpha^{1-\sigma}\right)}\right]^{\frac{1}{\sigma}}.$$

As in Section IV, budgetary persistence makes mandatory spending valuable, leading to overaccumulation of debt. When $\alpha$ is sufficiently large, the multiplier $\lambda$ becomes positive; the incumbent realizes that, despite whoever will be in power, she will enjoy the same level of mandatory spending. As a result, the incumbent internalizes the value of future resources, and debt starts decreasing.

Another important difference compared to Section IV is that in panel A of Figure 6, debt is strictly positive for all $\alpha \in [0,1]$. We obtain this result because party *I* does not fully commit period 2's budget. Debt is strictly below the sequestration threshold because party *I* wants to have some fiscal space to provide discretionary spending in case she is reelected. Since some risk remains, the cost of debt is not fully internalized, even when $g_0^O = 0$ and $\alpha = 1$.

As in the benchmark setting, panel B of Figure 6 shows that sequestration occurs when $g_0^O > 0$ and $\alpha$ is large enough. Note that there is now a discontinuous jump. Since triggering a sequestration has a sizable welfare cost (because discretionary spending is zero), triggering a "small" sequestration would be inefficient. Also note from panel B of Figure 5 that prior to triggering sequestration, mandatory spending is overprovided for dilution purposes. The first-order conditions under sequestration are derived in online Appendix C.

## C. *Political Polarization*

The analysis so far assumed extreme political polarization, wherein party *O* (party *I*) exclusively values goods *O* (goods *I*). The model can be easily extended to study the implications of varying levels of political polarization. We generalize utility (16) by assuming that party $j = I, O$ partially values goods $-j$. The utility function for party $j = I, O$ is represented by

$$(26) \qquad u_j\left(g^j, g^{-j}, d^j, d^{-j}\right) = \frac{\left(g^j\right)^{1-\sigma} + \omega\left(g^{-j}\right)^{1-\sigma}}{1-\sigma} + \theta\frac{\left(d^j\right)^{1-\sigma} + \omega\left(d^{-j}\right)^{1-\sigma}}{1-\sigma}.$$

As in Section VB, goods $d$ are nonsticky and can be cut at no cost by the current government, while goods $g$ are sticky, and $\theta \geq 0$. The parameter $\omega \in [0,1]$ measures preference alignment between the two parties. When $\omega = 0$, polarization is at its maximum, and the results coincide with those obtained so far. As $\omega$ increases, parties' preferences become more aligned, with each party starting to value the spending program preferred by the other party. Polarization converges toward zero as $\omega$ approaches one.

Under less than full polarization, there is the possibility that when party $j$ is in power, the "sticky" constraint $g_t^{-j} \geq \alpha g_{t-1}^{-j}$ may not be binding. The reason is straightforward: since party $j$ now values good $g^{-j}$, it could allocate additional resources to this good above and beyond prior commitments. Solving this model can be somewhat intricate due to the numerous possible cases, contingent on whether the sticky constraints for either the incumbent or the opposition, in either period 1 or period 2, may be slack. Nevertheless, as we discuss in online Appendix H, the results consistently align with intuitive expectations. First, as polarization diminishes, the incentives for excessive debt accumulation decrease. Similarly, lower polarization weakens the incentives for overproviding mandatory spending (sticky goods $g$) compared to discretionary spending (nonsticky goods $d$). Intuitively, when the two parties have more similar preferences, there is less need to use debt and sticky spending to strategically manipulate subsequent governments.

Overall, these results confirm the standard finding of the strategic debt literature that polarization is a driving force behind the increase in debt (see Yared 2019). In addition, our theory provides implications regarding the composition of government spending; higher polarization also implies an increase in sticky spending relative to nonsticky spending. As shown in Section VIC, these findings are generally in line with data from the United States, where over the past decades, we have witnessed the concurrent trends of increasing polarization, escalating debt, and an expanding proportion of mandatory spending within the budget.

## VI. Discussion

### A. *Sequestration Rules*

Several rules regulate US sequestration. In the online Appendix, we modify (6) to incorporate the two main features of the sequestration formula discussed in Section IIA. As summarized in this section, these adjustments do not alter the qualitative nature of the results in this paper. In fact, results might even be reinforced.

One feature of US sequestration is that cuts are equally divided between defense and nondefense programs. A consequence of this rule is that a relatively smaller percentage reduction is applied to the larger program (whether defense or nondefense). This implication was noted by White and Wildavsky (1989, p. 456): "Given that the balance of defense spending constituted 60 percent of non-excluded spending and only half the cut would be applied to defense, the sequestration percentage for defense programs would be lower than that applied to domestic programs." Since the larger program, whether in the defense or nondefense category, undergoes a lesser reduction, this exacerbates the dilution problem, creating a stronger incentive to increase spending (for more details, see online Appendix F).

The other main feature of the sequestration process is that certain programs, such as Social Security and Medicaid, are exempt from cuts. In online Appendix G, we model these exemptions and show that the results are similar to those in the baseline model. Sequestration exceptions, however, may affect politicians' incentives to allocate spending between exempt and nonexempt programs. If sequestration is not expected, we show that the ratio between these two programs remains undistorted compared to the first best. However, if sequestration is anticipated, our analysis shows that exempted programs may be overprovided or underprovided, depending on parameters. The origins of this ambiguity stem from the fact that sequestration exceptions make exempted programs more inertial than spending for nonexempted programs during sequestration. As we have discussed, higher spending inertia gives rise to income and substitution effects on spending incentives, explaining why sequestration exceptions may lead to either overprovision or underprovision of exempted programs.

## B. *Delayed Stabilization*

The previous analysis assumes that fiscal consolidation is automatically triggered when commitments become unsustainable. However, in practice, politicians are known for delaying fiscal adjustments until no further delay is possible. In the models so far, we have abstracted from these issues by assuming that entitlements can become unsustainable only in the second period, when delaying stabilization is not possible. To study delayed stabilization, in online Appendix D we assume that entitlements are unsustainable already at $t = 1$. We endow the period 1 incumbent with the choice of whether to sequester or leave it to the next government. Thus, the timing of fiscal consolidation becomes an endogenous outcome, and we establish conditions under which delay occurs.

Since entitlements inherited from period 0 determine how resources will be shared in a consolidation at time 1, we find that a higher ratio $g_0^O / g_0^I$ generally leads to a delay. Intuitively, a consolidation is postponed to time 2 if the party in power during period 1 has an unfavorable entitlements ratio, which she will try to improve by spending more on the goods she values. When a delay occurs, the incumbent meets existing commitments and funds additional spending by accumulating more debt, which makes the next-period consolidation even more painful. Besides the positive effect on current utility, the additional spending increases the relative size of her preferred programs, which dilutes the opposition's entitlements in a future consolidation. Still, we show that fiscal consolidation is not delayed if *total* entitlements are large enough. As total commitments increase, the fiscal space of the incumbent is reduced; after paying existing entitlements, she would have a small margin for improving the ratio, making delay a less profitable strategy.

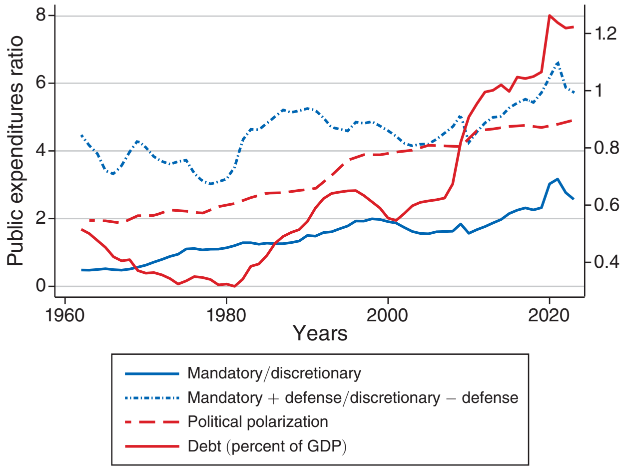


FIGURE 7. POLARIZATION, DEBT, AND SPENDING

Overall, these results suggest that in the United States, we will meet the conditions for significant entitlements reform by a government when accumulated spending commitments reach a sufficiently high level and when that government faces a favorable commitments ratio.

## C. *Empirical Implications*

In this section, we discuss whether the mechanisms of our model are consistent with the empirical data. Some findings are difficult to test. For example, testing Proposition 1 is challenging because it is difficult to infer the stickiness parameter from observed fiscal variables. Therefore, we focus on other implications of the model with more direct empirical counterparts.

As discussed in Section VC, our model implies that increased polarization can explain two potentially distinct phenomena observed in many economies: the rise in public debt and a shift in the composition of government spending away from nonsticky (discretionary) programs (for the United States, see Austin 2017 and Austin 2014).

In Figures 7 and 8, we illustrate the joint evolution of polarization, gross federal debt to GDP, and several spending ratios in the United States over the last few decades. As a measure of polarization, we consider the first dimension of DW-NOMINATE (see Lewis et al. 2023), which is designed to capture the "liberal" versus "conservative" distinction, plotting the score difference between the average Republican and Democratic members of the US House of Representatives.[28] As a proxy for the share of sticky spending over nonsticky spending, we calculate the ratio between mandatory and discretionary spending. In addition, due to the significant inertia observed in defense spending (see Section III), we create a time

[28] Azzimonti (2018) provides an alternative measure based on newspapers reporting political disagreement.

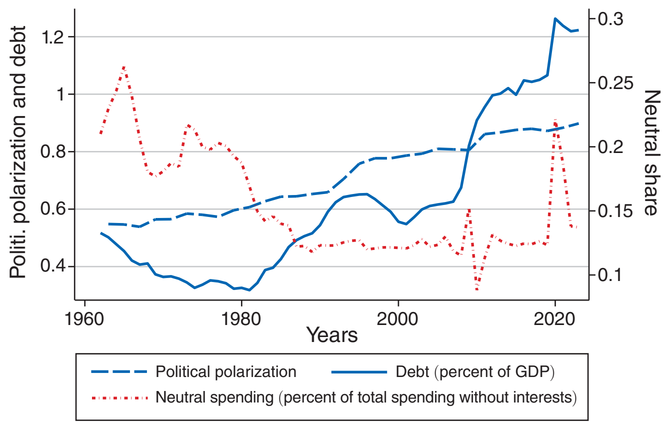


Figure 8. Neutral Spending

series by combining mandatory and defense spending, then dividing the total by the remaining items in the budget.

Figure 7 shows that political polarization positively correlates with debt over GDP, aligning with our model and the existing strategic debt literature. Moreover, rising political polarization is associated with a higher share of inertial spending, which is in line with the discussion in Section VC: increasingly polarized parties prioritize sticky (mandatory) programs that enable them to strategically manipulate future governments. We emphasize that while Figure 7 is consistent with our model, it is also consistent with other explanations that do not rely on increasing polarization. For example, it could result from aging voters demanding more inertial spending, such as health programs, akin to a decrease in $\theta$ in equation (26). One contribution of our model is to highlight that the share of inertial programs can grow even without an increase in voters' demand for mandatory spending, simply as a result of increasing polarization (a decrease in $\omega$). It bears stressing that these two explanations are not mutually exclusive; strategic considerations due to rising polarization might have amplified the effect due to higher voter demand.

In Figure 8, we show that increasing polarization is also associated with a decrease in the budget share of "neutral" spending, i.e., spending not "owned" by a party.[29] Applying a similar argument as before, a possible explanation is that bipartisan spending holds no value in terms of "strategic manipulation" since both parties agree on it. As a consequence, increasing polarization over partisan programs would lead to underprovision of neutral spending. Again, these strategic considerations may complement other explanations, such as reduced voters' demand for neutral spending.

Another prediction of our model is that parties may accelerate spending in the shadow of fiscal consolidation. To investigate this, we document the evolution of spending before the start year of the consolidation episodes obtained from Alesina,

[29] We categorize 32 OMB subfunctions as neutral following Table 1A in Epp, Lovett, and Baumgartner (2014b). After excluding "interest on the debt" from the analysis, the major neutral functions are transportation, general science, space, technology, and veterans' benefits and services. Note that this classification remains constant over time.

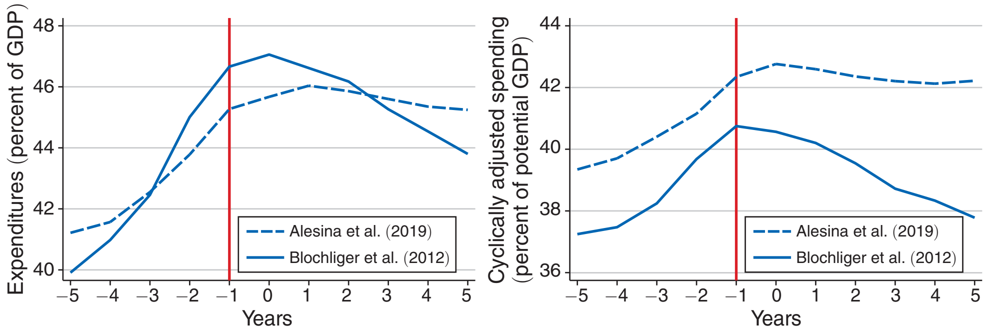


Figure 9. Spending Evolution before and during Consolidation

*Note:* Actual spending (left panel) and cyclically adjusted spending (right panel)

Favero, and Giavazzi (2019). In addition to these episodes, we also consider a set of 13 large consolidation episodes obtained from Blöchliger, Song, and Sutherland (2012), who selected them based on significant improvements in the primary balance.[30] Figure 9 shows on the left the evolution of average spending (as a percentage of GDP) five years prior to consolidation and during the consolidation for both Alesina, Favero, and Giavazzi (2019) and Blöchliger, Song, and Sutherland's (2012) episodes; on the right, we display cyclically adjusted spending (net of interest) as a percentage of potential GDP.[31] The figure shows that there is a steep rise in spending before consolidation, even after controlling for the economic cycle. The only difference between the two sets of episodes is that Alesina, Favero, and Giavazzi (2019) consolidations, not necessarily large, result in a lesser decrease in spending, as a percentage of GDP, during consolidation.

While Figure 9 is qualitatively consistent with our mechanism, it is econometrically challenging to disentangle it from another explanation that does not rely on strategic behavior; namely, a consolidation becomes more likely after years of fiscal irresponsibility. That being said, anecdotal evidence does suggest that policymakers are strategic, i.e., they recognize that spending is persistent (as seen in the quotes in the introduction) and that increasing spending can influence how resources will be distributed in future sequestration.

The following quote, which refers to the 2009 budget negotiations between Republicans and Democrats, aligns with the mechanism in our model. Specifically, it suggests that President Obama's stimulus package was a deliberate strategy, from a Democratic perspective, aimed at improving the allocation of resources in future consolidations:

[30] The episodes are Australia (1984–1988 and 1994–1998), Canada (1993–1997), Denmark (1983–1986), Finland (1993–2000), the United Kingdom (1993–1998), Greece (1990–1994), Ireland (1982–1988), Italy (1990–1995), Japan (1979–1987), Sweden (1981–1987 and 1994–1997), and the United States (1993–1998). As these episodes are less recent than those selected by Alesina, Favero, and Giavazzi (2019), COFOG data by spending function is available for only 2 out of the 13 episodes. In online Appendix I, we show that for these two episodes, both ex post and ex ante shares align along the diagonal, as in Figure 1.

[31] Since consolidations may follow each other closely in time, we exclude a consolidation episode that occurs within five years of another one.

> Both political parties are living in the shadow of the hard choices that are going to be imposed by the insolvency of America's entitlements. At some point soon, liberals are going to have to accept somewhat less spending than they'd like, and conservatives are going to have to accept somewhat higher taxes. And if you can change the baseline of social spending that Americans expect from their government before that day of hard choices arrives—and once created, government programs are awfully hard to get rid of, whether they're actually effective or not—then you've tilted the landscape of negotiation in liberalism's favor, and ensured that a post-Obama entitlement compromise will look a lot more like social democracy than a pre-Obama compromise would have.[32]

These strategic considerations may have led Democrats to advocate for higher spending than they would have otherwise. Of course, once Republicans came to power, they faced the same incentives Democrats had faced. President Trump's spending on defense, border control, and tax cuts may have been implemented with an eye toward the long-term effects of these policies, as a means of shifting back the balance in favor of Republicans.[33]

We stress that in the canonical strategic debt model of Alesina and Tabellini (1990), it would be difficult to rationalize that fiscal irresponsibility begets more irresponsibility. One would expect that higher spending, resulting in increased debt services, would act as a fiscal straitjacket, leading to more fiscal discipline. Following this argument, some commentators expected that Obama's increased spending would "force tax increases long after he had left office."[34] In the short run, the opposite occurred: fiscal indiscipline by one party triggered more irresponsibility in the other.

This pattern of "mutual fiscal indiscipline" does not appear to be limited to the Obama-Trump dynamics. For example, the empirical analysis by Romer and Romer (2009) finds that in the postwar period, following tax cut episodes, US federal spending doesn't decrease and may even increase after a few years. According to Romer and Romer (2009), these results provide evidence for what they term "shared fiscal irresponsibility," contradicting the "starve-the-beast" prediction that tax cuts force future policymakers to reduce spending. Similarly, Gale and Orszag (2004, p. 999) also emphasize that "abandoning fiscal discipline on one side of the budget could induce a period of fiscal irresponsibility on both sides of the budget." Our theoretical model provides a mechanism that allows us to rationalize this evidence.

Finally, a feature of our model is that large opposition commitments make a party less concerned about the deterioration of the fiscal balance (see equation (13)). In the presence of sticky spending, fiscal crises can serve as a means to dilute the opposition's entitlements (e.g., see "Blowing Up the Deficit Is Part of the Plan," *Atlantic*, December 19, 2017). In line with this mechanism, Republican voters are less inclined than Democrats to support government actions aimed at ensuring the financial stability of the Medicare system (see Egan 2013, Table 2.5). Similarly,

[32] Douthat, Ross. 2009. "The Pursuit of Social Democracy." *Atlantic*. March 4.

[33] For example, some commentators have argued that the substantial increase in defense spending approved in 2018 under Trump "sets a new baseline for future budget deals, and this could have long-lasting implications." (O'Brien, Connor. 2018. "Military hawks win big in budget deal—for now." *Politico*. February 9.)

[34] *Wall Street Journal*. 2017. "The Health Bill's Fiscal Bonus." March 14.

large Democratic entitlements may explain why Republican voters are less worried than Democrats about the consequences of breaking the debt ceiling.[35]

## VII. Conclusions

The unsustainability of current policies is the "fiscal problem of the 21st century" (Jones 2003). In the past two decades, bipartisan commissions have called for drastic entitlement-spending reforms. Politicians are not only not taking action, they seem to be following the opposite path; in recent years, their spending decisions have been worsening the US fiscal balance. It is possible that politicians underestimate the magnitude of the problem, but we provide another, possibly complementary, explanation. We show that when budgets are sticky, the prospect of stabilization makes politicians fiscally irresponsible; when debt is on an unsustainable path under current policies, politicians have the incentive to worsen the fiscal balance by spending even more to "dilute" existing entitlements.

Our model provides several insights on how to avoid this spending spiral. A widely held view is that budgetary inertia is responsible for high debt buildups. Due to the nontrivial impact of budgetary stickiness on the deficit bias, our results show that reducing budgetary inertia may worsen the debt problem. We show that when political institutions combine budgetary stickiness with a supermajority requirement for enacting policy changes, debt is reduced and, in some cases, eliminated. In the United States, Republicans and Democrats alike have passed several bills and tax reforms by using the so-called budget-reconciliation process, which allows the Senate to bypass the supermajority requirement. One conclusion of our analysis is that enforcing the Senate's 60 percent rule would lower US debt.

## Appendix

### A. *Proof of Proposition 1*

To prove part (i) of the proposition, we compute equilibrium debt when $\alpha = 0$ and when $\alpha = 1$. When $\alpha = 0$, a sequestration never occurs because (5) never holds. The first-order condition is then

$$(\tau + b_1)^{-\sigma} = (\tau - b_1)^{-\sigma} q, \tag{27}$$

which can be easily solved as follows:

$$b^*_{\alpha=0} = \frac{\tau\left(1 - q^{\frac{1}{\sigma}}\right)}{\left(1 + q^{\frac{1}{\sigma}}\right)}. \tag{28}$$

Next, assume $\alpha = 1$. We first show that party *I* does not choose a level of debt strictly below the sequestration threshold $\tilde{b}$. When $\alpha = 1$, recall that $\tilde{b} = 0$. To

[35] Pew Research Center (2013) and Cillizza, Chris, and Sean Sullivan. 2013. "Worried about the debt ceiling? Republicans aren't." *Washington Post*. October 8.

show that $b_1 < 0$ is not optimal, take the first-order condition (11) when $\alpha = 1$ under no sequestration:

$$(29) \qquad \left(\tau + b_1 - g_0^O\right)^{-\sigma} = \frac{q}{2-q}\left(\tau - b_1 - g_0^O\right)^{-\sigma}.$$

Since $q/(2-q) \leq 1$, we have that $b_1 < 0$ is not optimal. The first-order condition under sequestration (12) can be written as

$$(30) \qquad \left(\tau + b_1 - \alpha g_0^O\right)^{-\sigma} = \left[\left(\frac{\tau - b_1}{\tau + b_1}\right)\left(\tau + b_1 - \alpha g_0^O\right)\right]^{-\sigma} \times \left(1 - \frac{2\tau}{(\tau + b_1)^2}\alpha g_0^O\right).$$

After simplifying both sides, rewrite the above condition as

$$(31) \qquad (\tau + b_1)^{-\sigma} = (\tau - b_1)^{-\sigma}\left[1 - \frac{2\tau}{(\tau + b_1)^2}\alpha g_0^O\right].$$

Since the right-hand side in (31) is increasing in $b_1$, while the left-hand side is decreasing, the solution is unique. For example, as $\sigma \to 1$ and $\alpha = 1$, we obtain a simple closed-form solution for $b_1$, given by (15).

To show that $b^*_{\alpha=0} < b^*_{\alpha=1}$, we need to show that when (31) is evaluated at $b^*_{\alpha=0}$, we have

$$(32) \qquad (\tau + b^*_{\alpha=0})^{-\sigma} > (\tau - b^*_{\alpha=0})^{-\sigma}\left[1 - \frac{2\tau}{(\tau + b^*_{\alpha=0})^2}\alpha g_0^O\right].$$

That is, the marginal benefit of debt is higher than the marginal cost of debt, implying that optimal debt is higher than $b^*_{\alpha=0}$. Focusing on $\alpha = 1$, and using the first-order condition (27), the inequality above can be written as

$$(33) \qquad \left[1 - \frac{2\tau}{(\tau + b^*_{\alpha=0})^2} g_0^O\right] < q.$$

Plugging (28), we obtain

$$(34) \qquad \left[1 - \frac{\left(1 + q^{\frac{1}{\sigma}}\right)^2}{2\tau} g_0^O\right] < q$$

or

$$(35) \qquad g_0^O > \frac{2\tau(1-q)}{\left(1 + q^{\frac{1}{\sigma}}\right)^2}.$$

This condition implies that $b^*_{\alpha=0} < b^*_{\alpha=1}$, as stated in Proposition 1.

We now prove the second part of Proposition 1. We first show that the partial derivative of equilibrium debt with respect to $\alpha$ is positive when evaluated at $\alpha = 0$. When $\alpha$ is close to zero, sequestration is not triggered. One can thus compute the interior solution for debt as follows:

$$b_1^* = \frac{\left(\left[\frac{1 + \alpha^{1-\sigma}(1-q)}{q}\right]^{\frac{1}{\sigma}} - 1\right)\tau - \left[\alpha\left(\frac{1 + \alpha^{1-\sigma}(1-q)}{q}\right)^{\frac{1}{\sigma}} - 1\right]\alpha g_0^O}{\left[\frac{1 + \alpha^{1-\sigma}(1-q)}{q}\right]^{\frac{1}{\sigma}} + 1}. \tag{36}$$

We let $D$ and $N$ denote, respectively, the denominator and the numerator in (36). Moreover, we define $\Delta \equiv \left[1 + \alpha^{1-\sigma}(1-q)\right]$.

$$\frac{\partial b_1^*}{\partial \alpha} = \frac{1}{D^2}\Big[(\tau D - N)\,\alpha^{-\sigma}\frac{(1-\sigma)(1-q)}{\sigma}\Delta^{\left(\frac{1}{\sigma}-1\right)} + g_O q^{1/\sigma} D - D2\alpha g_0^O \Delta^{(1/\sigma)} - \alpha^{2-\sigma} g_O \frac{1-\sigma}{\sigma}\Delta^{\left(\frac{1}{\sigma}-1\right)}(1-q)D\Big] \tag{37}$$

When $\alpha$ is close to zero, we have $\tau D > N$. Then, the limit of (37) is $+\infty$ as $\alpha \to 0$ provided that $\sigma < 1$. Therefore, equilibrium debt is increasing in $\alpha$ when $\alpha$ is sufficiently close to zero. When instead $\alpha$ is close to one and, consequently, sequestration is triggered, from (31), equilibrium debt is increasing in $\alpha$ when $g_0^O > 0$. When $\alpha = 1$ and $g_0^O = 0$, equilibrium debt is zero. This discussion proves that equilibrium debt is a nonmonotone function of $\alpha$. ∎

## B. *Proof of Proposition 2*

Assume $\alpha = 1$ and let the initial entitlements $\left(g_0^O, g_0^I\right)$ be given. We will first show (Step 1) that among all debt levels that trigger a sequestration, zero debt is optimal. We will then establish (Step 2) that the period 1 incumbent finds it optimal to trigger a sequestration in period 2.

**Step 1:** Zero debt is the optimal policy among all policies that result in a sequestration in period 2.

Assume that party $I$ chooses $b_1 \geq \tilde{b}$, so that a sequestration will occur in the final period. We will verify that this assumption is correct later in Step 2. Since $\alpha = 1$, the sequestration threshold is $\tilde{b} = 0$ and, consequently, sequestration will occur when $b_1 \geq 0$. We define the variable $\chi \equiv (\tau - b_1)/(\tau + b_1)$. When $b_1 \geq 0$, we have $\chi \leq 1$. The agenda setter's problem can be written as

$$\max_{\left\{b_1, g_1^O, g_1^I\right\}} u\left(g_1^I\right) + u\left(\chi g_1^I\right)$$

subject to

$$u\left(g_1^O\right) + u\left(\chi g_1^O\right) \geq u\left(g_0^O\right) + qu\left(g_0^O\right) + (1-q)u\left(2\tau - 2g_0^I - g_0^O\right) \tag{AC}$$

and

$$g_1^I + g_1^O = \tau + b_1.$$

Let $\lambda$ be the multiplier associated with the acceptance constraint. Taking derivatives with respect to $g_1^O$, we obtain

$$u'(\tau + b_1 - g_1^O) + \chi u'(\chi g_1^I) = \lambda\left[u'(g_1^O) + \chi u'(\chi g_1^O)\right]$$

or

$$u'(g_1^I)(1 + \chi^{1-\sigma}) = \lambda\left[u'(g_1^O)(1 + \chi^{1-\sigma})\right].$$

For this equation to hold,

$$\lambda = \frac{u'(\tau + b_1 - g_1^O)}{u'(g_1^O)}. \tag{38}$$

Next, we take the derivative with respect to $b_1$ and obtain

$$\begin{aligned} 0 = & -\lambda\left[u'(\chi g_1^O) g_1^O \frac{2\tau}{(\tau + b_1)^2}\right] + u'(\tau + b_1 - g_1^O) \\ & + u'\left(\chi(\tau + b_1 - g_1^O)\right)\left[\frac{-2\tau}{(\tau + b_1)^2}(\tau + b_1 - g_1^O) + \frac{(\tau - b_1)}{\tau + b_1}\right]. \end{aligned} \tag{39}$$

After inserting (38), we verify that when $b_1 = 0$, the above condition is satisfied. In fact,

$$u'(\tau - g_1^O) + u'(\tau - g_1^O)\left[\frac{-2}{\tau}(\tau - g_1^O) + 1\right] - \frac{2u'(\tau - g_1^O)}{\tau} g_1^O = 0 \tag{40}$$

or

$$u'(\tau - g_1^O) = u'(\tau - g_1^O). \tag{41}$$

Zero debt is, therefore, optimal. Note that $g_1^O$ solves the acceptance constraint with equality given the initial default option $(g_0^O, g_0^I)$. That is,

$$2u(g_1^O) = u(g_0^O) + qu(g_0^O) + (1 - q)u(2\tau - 2g_0^I - g_0^O). \tag{42}$$

**Step 2:** We show that leaving fiscal space for the subsequent government is not optimal for the period 1 incumbent. In other words, choosing $b_1 < \tilde{b} = 0$ is not optimal.

When $\alpha = 1$, the sequestration threshold is $\tilde{b} = 0$. When there is no sequestration, political uncertainty in the second period is maintained. The agenda setter's problem is written as follows:

$$\max_{\{b_1, g_1^O, g_1^I\}} u(\tau + b_1 - g_1^O) + qu(\tau - b_1 - g_1^O) + (1-q)u(\tau + b_1 - g_1^O)$$

subject to

$$u(g_1^O) + qu(g_1^O) + (1-q)u(g_1^O - 2b_1) \geq u(g_0^O) + qu(g_0^O) + (1-q)u(2\tau - 2g_0^I - g_0^O).$$

We take the derivatives with respect to $g_1^O$:

$$(43) \qquad -u'(\tau + b_1 - g_1^O)(2-q) - qu'(\tau - b_1 - g_1^O) + \lambda\left[u'(g_1^O)(1+q) + u'(g_1^O - 2b_1)(1-q)\right] = 0.$$

We take the derivatives with respect to $b_1$:

$$(44) \qquad u'(\tau + b_1 - g_1^O)(2-q) - qu'(\tau - b_1 - g_1^O) - \lambda\left[(1-q)u'(g_1^O - 2b_1)2\right] = 0.$$

We evaluate the first-order condition (43) at $b_1 = 0$:

$$(45) \qquad \lambda = \frac{u'(\tau - g_1^O)(1-q)}{u'(g_1^O)}.$$

The first-order condition (44) evaluated at $b_1 = 0$ does not hold. In fact,

$$(46) \qquad u'(\tau - g_1^O)(2-q) - qu'(\tau - g_1^O) - \frac{u'(\tau - g_1^O)(1-q)}{u'(g_1^O)}\left[2(1-q)u'(g_1^O)\right] > 0.$$

Thus, the two parties fully commit all the resources in the second period: $\chi = 1$. ∎

## REFERENCES


**Alesina, Alberto, Gualtiero Azzalini, Carlo A. Favero, Francesco Giavazzi, and Armando Miano.** 2016. "Is it the "How" or the "When" that Matters in Fiscal Adjustments?" Unpublished.

**Alesina, Alberto, and Allan Drazen.** 1991. "Why Are Stabilizations Delayed?" *American Economic Review* 81 (5): 1170–88.

**Alesina, Alberto, Carlo Favero, and Francesco Giavazzi.** 2019. *Austerity: When It Works and When It Doesn't*. Princeton, NJ: Princeton University Press.

**Alesina, Alberto, and Andrea Passalacqua.** 2015. "The Political Economy of Government Debt." NBER Working Paper 21821.

**Alesina, Alberto, and Francesco Passarelli.** 2019. "Loss Aversion in Politics." *American Journal of Political Science* 63 (4): 936–47.

**Alesina, Alberto, and Guido Tabellini.** 1990. "A Positive Theory of Fiscal Deficits and Government Debt." *Review of Economic Studies* 57 (3): 403–14.

**Arellano, Cristina.** 2008. "Default Risk and Income Fluctuations in Emerging Economies." *American Economic Review* 98 (3): 690–712.

**Auerbach, Alan J., Jagadeesh Gokhale, and Laurence J. Kotlikoff.** 1994. "Generational Accounting: A Meaningful Way to Evaluate Fiscal Policy." *Journal of Economic Perspectives* 8 (1): 73–94.

**Austin, D. Andrew.** 2014. *The Budget Control Act and Trends in Discretionary Spending*. Washington, DC: Congressional Research Service.

**Austin, D. Andrew.** 2017. *Present Trends and the Evolution of Mandatory Spending*. Washington, DC: Congressional Research Service.

**Azzimonti, Marina.** 2018. "Partisan Conflict and Private Investment." *Journal of Monetary Economics* 93 (C): 114–31.

**Azzimonti, Marina, Laura Karpuska, and Gabriel Mihalache.** 2023. "Bargaining over Taxes and Entitlements in the Era of Unequal Growth." *International Economic Review* 64 (3): 893–941.

**Balog, Matej, Nilesh Tripuraneni, Zoubin Ghahramani, and Adrian Weller.** 2017. "Lost Relatives of the Gumbel Trick." In *Proceedings of the 34th International Conference on Machine Learning*, Vol. 70, edited by Doina Precup and Yee Whye Teh, 371–79. Sydney, Australia: PMLR.

**Baron, David P.** 1996. "A Dynamic Theory of Collective Goods Programs." *American Political Science Review* 90 (2): 316–30.

**Battaglini, Marco, and Stephen Coate.** 2008. "A Dynamic Theory of Public Spending, Taxation, and Debt." *American Economic Review* 98 (1): 201–36.

**Blöchliger, Hansjörg, Dae-Ho Song, and Douglas Sutherland.** 2012. "Fiscal Consolidation: Part 4. Case Studies of Large Fiscal Consolidation Episodes." Unpublished.

**Bohn, Henning.** 2005. "The Sustainability of Fiscal Policy in the United States." Unpublished.

**Bolton, Patrick, and Olivier Jeanne.** 2009. "Structuring and Restructuring Sovereign Debt: The Role of Seniority." *Review of Economic Studies* 76 (3): 879–902.

**Bouton, Laurent, Alessandro Lizzeri, and Nicola Persico.** 2020. "The Political Economy of Debt and Entitlements." *Review of Economic Studies* 87 (6): 2568–99.

**Bove, Vincenzo, Georgios Efthyvoulou, and Antonio Navas.** 2017. "Political Cycles in Public Expenditure: Butter vs Guns." *Journal of Comparative Economics* 45 (3): 582–604.

**Bowen, T. Renee, Ying Chen, and Hülya Eraslan.** 2014. "Mandatory versus Discretionary Spending: The Status Quo Effect." *American Economic Review* 104 (10): 2941–74.

**CBO.** 2020. *The Budget and Economic Outlook: 2020 to 2030*. Washington, DC: Congressional Budget Office.

**Charité, Jimmy, Raymond Fisman, and Ilyana Kuziemko.** 2015. "Reference Points and Re-distributive Preferences: Experimental Evidence." NBER Working Papers 21009.

**Chatterjee, Satyajit, and Burcu Eyigungor.** 2012a. "Debt Dilution and Seniority in a Model of Defaultable Sovereign Debt." Unpublished.

**Chatterjee, Satyajit, and Burcu Eyigungor.** 2012b. "Maturity, Indebtedness, and Default Risk." *American Economic Review* 102 (6): 2674–99.

**Chatterjee, Satyajit, and Burcu Eyigungor.** 2020. "Policy Inertia, Election Uncertainty, and Incumbency Disadvantage of Political Parties." *Review of Economic Studies* 87 (6): 2600–2638.

**Cruz, Cesi, Philip Keefer, and Carlos Scartascini.** 2018. *Database of Political Institutions 2017.* Washington, DC: Inter-American Development Bank Research Department. https://mydata.iadb.org/Reform-Modernization-of-the-State/Database-of-Political-Institutions-2017/4umk-88xe.

**Davis, Otto A., M. A. H. Dempster, and Aaron Wildavsky.** 1966. "A Theory of the Budgetary Process." *American Political Science Review* 60 (3): 529–47.

**Debortoli, Davide, and Ricardo Nunes.** 2013. "Lack of Commitment and the Level of Debt." *Journal of the European Economic Association* 11 (5): 1053–78.

**Dempster, M. A. H., and Aaron Wildavsky.** 1979. "On Change: Or, There is No Magic Size for An Increment." *Political Studies* 27 (3): 371–89.

**Egan, Patrick J.** 2013. *Partisan Priorities: How Issue Ownership Drives and Distorts American Politics*. Cambridge: Cambridge University Press.

**Epp, Derek A., John Lovett, and Frank R. Baumgartner.** 2014a. "Partisan Priorities and Public Budgeting." *Political Research Quarterly* 67 (4): 864–78.

**Epp, Derek A., John Lovett, and Frank R. Baumgartner.** 2014b. "Supplementary Material for 'Partisan Priorities and Public Budgeting'." Unpublished.

**Eraslan, Hülya, and Adriana Piazza.** 2020. "Efficiency with Political Power Dynamics and Costly Policy Change." Unpublished.

**Eraslan, Hülya, Kirill Evdokimov, and Jan Zapal.** 2020. "Dynamic Legislative Bargaining." Unpublished.

**Gale, William G., and Peter R. Orszag.** 2004. "Bush Administration Tax Policy: Starving the Beast?" *Tax Notes* 105 (5): 999–1002.

**Gersbach, Hans, Matthew O. Jackson, Philippe Muller, and Oriol Tejada.** 2023. "Electoral Competition with Costly Policy Changes: A Dynamic Perspective." *Journal of Economic Theory* 214: 105716.

**Gersbach, Hans, Philippe Muller, and Oriol Tejada.** 2019. "Costs of Change and Political Polarization." *European Journal of Political Economy* 60 (C): 101801.

**Gordon, Grey.** 2019. "Efficient Computation with Taste Shocks." Unpublished.

**Hamilton, James D.** 2013. "Off-Balance-Sheet Federal Liabilities." NBER Working Paper 19253.

**Hatchondo, Juan Carlos, Leonardo Martinez, and Cesar Sosa-Padilla.** 2016. "Debt Dilution and Sovereign Default Risk." *Journal of Political Economy* 124 (5): 1383–1422.

**Jones, Bryan D., James L. True, and Frank R. Baumgartner.** 1997. "Does Incrementalism Stem from Political Consensus or from Institutional Gridlock?" *American Journal of Political Science* 41 (4): 1319–39.

**Jones, Charles I.** 2003. "The Fiscal Problem of the 21st Century." Unpublished.

**Konigsberg, Charles S.** 2019. *The Annual Sequester of Mandatory Spending through FY2029*. Washington, DC: Congressional Research Service.

**Krehbiel, Keith.** 1998. *Pivotal Politics: A Theory of U.S. Lawmaking*. Chicago, IL: University of Chicago Press.

**Lambertini, Luisa.** 2003. "Are Budget Deficits Used Strategically?" Unpublished.

**Lewis, Jeffrey B., Keith Poole, Howard Rosenthal, Adam Boche, Aaron Rudkin, and Luke Sonnet.** 2023. "Voteview: Congressional Roll-Call Votes Database." Vote View. https://voteview.com/ (accessed spring 2023).

**Lizzeri, Alessandro.** 1999. "Budget Deficits and Redistributive Politics." *Review of Economic Studies* 66 (4): 909–28.

**Lockwood, Ben, Apostolis Philippopoulos, and Andy Snell.** 1996. "Fiscal Policy, Public Debt Stabilisation and Politics: Theory and UK Evidence." *Economic Journal* 106 (437): 894–911.

**Lynch, Megan S.** 2011. *Statutory Budget Controls in Effect Between 1985 and 2002*. Washington, DC: Congressional Research Service.

**Lynch, Megan S.** 2013. *Sequestration as a Budget Enforcement Process: Frequently Asked Questions*. Washington, DC: Congressional Research Service.

**McFadden, Daniel.** 1973. "Conditional Logit Analysis of Qualitative Choice Behaviour." In *Frontiers in Econometrics*, edited by P. Zarembka, 105–42. New York, NY: Academic Press New York.

**OECD.** 2010. *Tax Expenditures in OECD Countries*. Paris: OECD.

**OECD.** 2012. *Recommendation of the Council on Regulatory Policy and Governance*. Paris: OECD.

**OECD.** 2021. *Government at a Glance*. Paris: OECD.

**OECD.** 2023. *General Government Spending* (*Indicator*). https://www.oecd.org/en/data/indicators/general-government-spending.html (accessed September 27, 2022).

**OECD.** 2024. *Gross Domestic Product* (*GDP*) (*indicator*). https://doi.org/10.1787/4537dc58en.

**OMB.** 2024. *Historical Tables*. https://www.whitehouse.gov/omb/budget/historical-tables/ (accessed September 1, 2023).

**Osborne, Martin J., and Ariel Rubinstein.** 1990. *Bargaining and Markets*. Bingley, UK: Emerald Group Publishing.

**Persson, Torsten, and Lars E. O. Svensson.** 1989. "Why a Stubborn Conservative would Run a Deficit: Policy with Time-Inconsistent Preferences." *Quarterly Journal of Economics* 104 (2): 325–45.

**Pew Research Center.** 2013. *Partisans Dug in on Budget, Health Care Impasse*. Washington, DC: Pew Research Center.

**Pew Research Center.** 2019. *Little Public Support for Reductions in Federal Spending*. Washington, DC: Pew Research Center.

**Piguillem, Facundo, and Alessandro Riboni.** 2021. "Fiscal Rules as Bargaining Chips." *Review of Economic Studies* 88 (5): 2439–78.

**Piguillem, Facundo, and Alessandro Riboni.** 2024. *Data and Code for:* "*Sticky Spending, Sequestration, and Government Debt.*" Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. https://doi.org/10.3886/E201443V1.

**Potrafke, Niklas.** 2017. "Partisan Politics: Empirical Evidence from OECD Panel Studies." *Journal of Comparative Economics* 45 (4): 712–50.

**Romer, Christina D., and David H. Romer.** 2009. “Do Tax Cuts Starve the Beast? The Effect of Tax Changes on Government Spending.” *Brookings Papers on Economic Activity* 39 (1): 139–214.

**Rust, John.** 1987. “Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher.” *Econometrica* 55 (5): 999–1033.

**Schick, Allen.** 2003. “The Role of Fiscal Rules in Budgeting.” *OECD Journal on Budgeting* 3 (3): 7–34.

**Stith, Kate.** 1988. “Rewriting the Fiscal Constitution: The Case of Gramm-Rudman-Hollings.” *California Law Review* 76 (3): 593–668.

**Tabellini, Guido, and Alberto Alesina.** 1990. “Voting on the Budget Deficit.” *American Economic Review* 80 (1): 37–49.

**Tanzi, Vito.** 2008. “The Future of Fiscal Federalism.” *European Journal of Political Economy* 24 (3): 705–12.

**White, Joseph, and Aaron Wildavsky.** 1989. *The Deficit and the Public Interest: The Search for Responsible Budgeting in the 1980s*. Berkeley, CA: University of California Press.

**Wildavsky, Aaron.** 1964. *The Politics of the Budgetary Process*. Toronto: Little, Brown and Co.

**Yared, Pierre.** 2010. “Politicians, Taxes and Debt.” *Review of Economic Studies* 77 (2): 806–40.

**Yared, Pierre.** 2019. “Rising Government Debt: Causes and Solutions for a Decades-Old Trend.” *Journal of Economic Perspectives* 33 (2): 115–40.